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Estimating Wind Speed availability by using AI method

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UNDERGRADUATE GRADUATION PROJECT APPROVAL FORM

The undergraduate graduation project titled Estimating Wind Speed availability by using AI method, prepared by LOUKMAN OMAR WAIS and ALOU COULIBALY under the supervision of Prof. Dr. İsmail Hakkı ALTAŞ was examined and accepted as an Undergraduate Graduation Project in terms of its scope and quality.

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ALOU COULIBALY

LOUKMAN OMAR WAIS

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SUMMARY

This study explores the application of artificial intelligence (AI) techniques for wind speed forecasting in Istanbul, Turkey, as a critical component of enhancing renewable energy integration within microgrids. Historical wind speed data, along with meteorological variables such as temperature, humidity, surface pressure, wind direction, wind gust and terrestrial radiation, were sourced from Open-Meteo and preprocessed through cleaning, outlier removal, normalization, and feature selection to ensure data quality and consistency.

A diverse set of machine learning algorithms, including Decision Tree, Linear Regression, Support Vector Machine (SVM), Neural Network, Efficient Linear Model, and Ensemble Methods, were developed and evaluated for predictive performance. Among these, the SVM model demonstrated the highest accuracy and reliability, emerging as the best-performing model based on key evaluation metrics.

The modeling and validation process was conducted in MATLAB, utilizing its advanced toolsets for data analysis and model optimization. Performance was assessed using Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2), with the SVM model achieving an RMSE of 0.8276 and an R^2 of 0.99, indicating high prediction precision. Other models, such as Ensemble Methods and Neural Networks, also yielded strong results, further validating the robustness of the methodology.

This study underscores the value of AI-driven approaches in renewable energy forecasting and supports data-informed decision-making for sustainable energy management. By combining advanced machine learning models with rigorous evaluation techniques, the research contributes to the optimization of wind energy utilization in urban microgrid systems.

SYMBOLS AND ABBREVIATIONS

AI	: Artificial Intelligence
ML	: Machine Learning
ANN	: Artificial Neural Network
MSE	: Mean Squared Error
MAE	: Mean Absolute Error
RMSE	: Root Mean Squared Error
R ²	: Coefficient of Determination
CSV	: Comma Separated Values
°C	: Degrees Celsius
RH	: Relative Humidity
LSTM	: Long Short-Term Memory
RNN	: Recurrent Neural Network
SVM	: Support Vector Machine
ARIMA	: Autoregressive Integrated Moving Average
RL	: Reinforcement Learning
m/s	: Meters per Second
%	: Percent
Pa	: Pascal
W/m ²	: Watt per square meter
Sin	: Sine
Cos	: Cosine
DL	: Deep Learning
°	: Degree
L	: Lider

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CHAPTER 1 : INTRODUCTION

1.1. General information

In the past decade, Conventional energy has been the most important energy source, therefore this energy is the cause of many problems such as global warming and increased greenhouse gas emissions. Wind energy is one of the most promising and sustainable sources of renewable energy. The evolution of wind energy is significant in this world for instance

The availability of wind energy is determined by the wind speed, which is not only critical for the design of wind turbines but also for evaluating the efficiency and feasibility of wind energy projects. Accurately estimating wind speed and predicting its availability is essential for optimizing the operation of wind farms and improving the efficiency of energy production.

In recent years, Artificial Intelligence (AI) techniques have been increasingly applied in the field of renewable energy, especially in the estimation and forecasting of wind speed. These techniques offer several advantages over traditional methods, such as improved accuracy and the ability to learn from historical data to make more reliable predictions. The primary goal of this project is to explore how AI methods, specifically machine learning algorithms, can be used to estimate wind speed availability and enhance the management of wind energy production.

In this study, we focus on utilizing AI techniques to analyze and predict wind speed patterns by leveraging historical wind data. The application of AI models aims to provide more accurate wind speed forecasts, enabling better resource management and supporting the planning and maintenance of wind turbines.

1.2. Literature Review

In this thesis study, comprehensive literature has been examined in detail and many studies have been used.

In the study conducted by Kusiak et al. (2011), machine learning techniques such as Support Vector Machines (SVM), Random Forests, and Artificial Neural Networks (ANN) were applied for wind speed forecasting. Similar to our research, where machine learning methods are employed for forecasting, their study aimed to improve the accuracy of predictions compared to traditional methods. However, the study highlighted challenges in modeling the temporal dynamics of wind data, where past observations strongly influence future outcomes. Despite these challenges, the study contributed to exploring more advanced machine learning

techniques in wind speed forecasting, offering valuable insights for improving prediction accuracy in this field.[1]

Hochreiter and Schmidhuber carried out a research on introduced Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, as a promising solution for time-series forecasting.[2].

In a more recent study, conducted by Yang et al. (2020) explored the integration of additional meteorological parameters such as temperature, pressure, and humidity into wind speed forecasting models. By combining these factors with wind speed data, the study found improved prediction accuracy.[3]

In their research publish by J. J. Hox and H. R. Boeije explain the difference between primary and secondary data collection, highlighting that primary data is collected directly by the researcher, while secondary data is reused from existing sources. They emphasize choosing the appropriate type based on the research purpose and data availability.[4]

The advancement of intelligent algorithms such as Artificial Neural Networks (ANNs) and Support Vector Machines (SVMs) has greatly influenced the accuracy and development of wind energy forecasting. In a study by Marugán et al. (2018), the use of ANNs in wind energy systems was examined, though it offered limited analysis of forecasting methodologies[5]. On the other hand, Zendehboudi et al. (2018) conducted a more detailed investigation into the application of SVMs for both solar and wind energy forecasting, demonstrating their strength in managing complex, non-linear data relationship.[6]s.

Hochreiter & Schmidhuber (1997) introduced Long Short-Term Memory (LSTM) networks, a type of recurrent neural network, as a promising solution for time-series forecasting. Their study demonstrated that LSTMs effectively capture long-term dependencies and non-linear dynamics, making them superior for wind speed forecasting.[7].

Although these studies demonstrate the effectiveness of machine learning and AI models in wind speed forecasting, most of them remain isolated in their application. Many focus on theoretical or small-scale testing and do not address the full potential of AI models for real-world applications, such as integration into wind farm operations and microgrid management. This project aims to fill this gap by developing a practical, deployable AI model that not only predicts wind speed accurately but also optimizes renewable energy system management in

real-time, thereby enhancing operational efficiency and reducing dependence on non-renewable energy sources.

1.3. Novelty

This study introduces a novel approach to wind speed forecasting by conducting a comprehensive comparison of multiple machine learning models using real meteorological data from Istanbul. Unlike conventional studies that often rely on a single predictive technique, this research evaluates six different algorithms: Decision Tree, Linear Regression, Support Vector Machine (SVM), Neural Network, Efficient Linear Model, and Ensemble Methods to determine the most accurate model for short-term wind speed prediction.

A key innovation of this project lies in the integration of a wide range of meteorological variables, including temperature, humidity, surface pressure, wind direction, wind gusts, and terrestrial radiation. This multi-variable approach enhances the learning capability of the models by capturing complex environmental patterns that influence wind behavior.

Furthermore, the implementation and validation of these models in MATLAB using its Machine Learning Toolbox provides a practical framework for real-world deployment in wind energy systems. This enables direct application of the results in microgrid optimization and operational planning for renewable energy systems.

By combining a rich dataset, a multi-model evaluation strategy, and practical MATLAB-based deployment, this project delivers a robust and scalable solution for accurate wind speed forecasting, setting it apart from traditional forecasting approaches in both methodology and applicability.

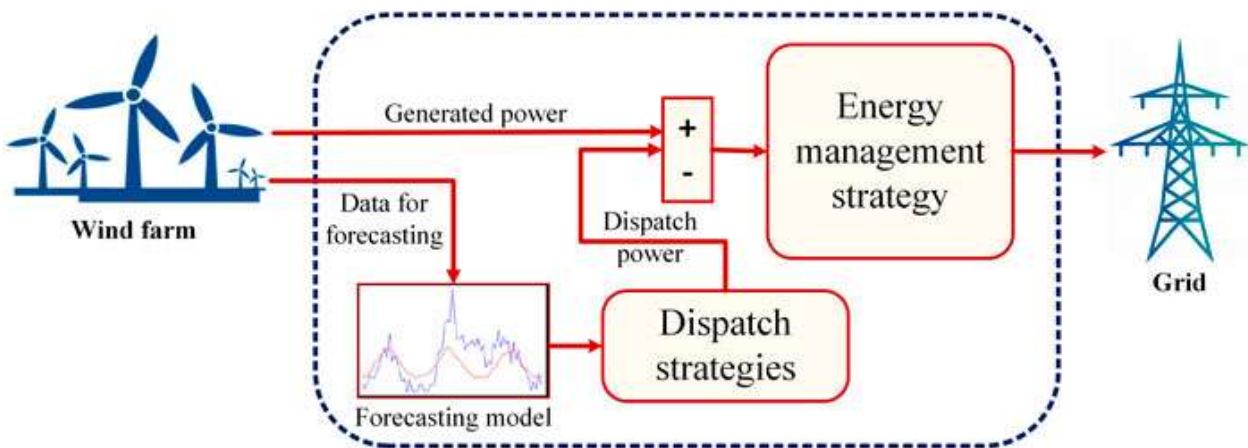
1.4. Methodology

Wind energy is a key component in the shift toward renewable energy sources, offering a sustainable alternative to fossil fuels. However, the performance of wind power systems depends on the availability and predictability of wind speed. Accurate forecasting is crucial for ensuring reliable electricity generation and effective microgrid management. In this study, wind speed prediction is approached through an AI-based methodology consisting of three main stages: data preparation, model development, and implementation using MATLAB.

The **first step** in the methodology involves data collection and preprocessing. Historical wind speed data is gathered alongside several meteorological variables, including temperature, humidity, wind speed at 10 m and 100 m, terrestrial radiation, wind direction at 10 m, surface pressure, and wind gusts at 10 m. The dataset undergoes thorough cleaning, which includes handling missing values, removing outliers, and selecting the most relevant features through feature engineering. The processed data is then normalized to ensure consistency and is divided into training and testing sets to prepare for model development.

The **second step** focuses on developing predictive models using various Machine Learning (ML) algorithms. These include Decision Tree, Linear Regression, Support Vector Machine (SVM), Neural Network, Efficient Linear Model, and Ensemble Methods. Each model is trained on the prepared dataset and evaluated for its predictive performance. The evaluation is guided by key error metrics, and through this process, the SVM model demonstrates superior accuracy and reliability, emerging as the most effective predictor among the tested algorithms.

The **third step** consists of implementing and validating the models using MATLAB. MATLAB serves as a robust platform for conducting advanced data analysis, model training, and performance evaluation. Using its comprehensive toolsets, the developed models are tested and optimized. Essential performance metrics such as Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and R-squared (R^2) are calculated to measure the prediction accuracy. This final evaluation stage enables a detailed comparison between models, supporting the selection of the most effective algorithm. The study concludes with the identification of the best-performing model and the presentation of validated experimental results.



As illustrated in **Figure 1.4**, the forecasting model is integrated into a larger energy management strategy. The wind farm generates power and provides data for forecasting.

This data feeds a forecasting model that predicts future wind availability. The predicted power is then passed to dispatch strategies, which determine how best to use or store the energy. The result is managed by an energy strategy system that ensures efficient and reliable delivery to the electrical grid.

1.5. Impact

The main impact of this project lies in its ability to significantly improve the efficiency and reliability of wind farm operations and microgrid systems. By providing accurate short term wind speed predictions, the AI-based model can help energy managers optimize energy production, anticipate periods of low or high wind speeds, and make informed decisions about energy storage and distribution. This would lead to more stable, efficient, and cost-effective management of renewable energy systems.

Additionally, this project can contribute to addressing the key challenges faced by renewable energy systems such as variability and intermittency by enhancing wind forecasting accuracy. As a result, the dependency on traditional energy sources, especially fossil fuels, can be reduced, supporting the transition to a cleaner, more sustainable energy system. The methodologies and techniques used in this study could also be applied to other renewable energy sources, thereby broadening the scope and potential impact.

Moreover, the deployment of AI and machine learning in the renewable energy sector can encourage the adoption of more advanced, data-driven approaches to energy management. By showcasing the ability of AI to improve prediction accuracy in such a complex domain, this project also has the potential to inspire further research and innovation in AI applications within the field of renewable energy.

1.6. Standards

Standards on wind speed for wind power are important for ensuring the reliable design, installation and operation wind energy systems.

- **IEC 61400-1:2019:** Wind energy generation-Design requirements.
- **IEC 61400-12-1:2022:** Wind energy generation systems - Power performance measurements of electricity producing wind turbines.
- **IEC 61400-21-1:2019:** Wind energy generation systems -Measurement and assessment of electrical characteristics - Wind turbines.
- **ISO 2533:1975:** Standard Atmosphere.

- **WMO-No. 8:** Guide to Meteorological Instruments.
- **IEEE 1159-2019:** Power Quality Monitoring

1.7. Work Schedule

Table 1.1 shows the step-by-step process of the Engineering Design and Graduation Project using a timeline. The project is divided into smaller work sections, each with its own tasks and deadlines. These time plans help keep the work going smoothly. To prepare for any possible problems, a backup plan “Plan B” has also been created. The risks and their solutions are explained in Table 1.2, which helps make sure the project can continue even if issues come up.

Table 1.1. Work-Time Table

WP No	Name and Purpose of the Work Package	Students Responsible	Time Range	Contribution to the Project
1	- Developing the project concept - Conducting an extensive literature review on wind energy forecasting, AI methods and existing models - Presenting the unique idea and justifying its originality with references from literature - Identifying the key objectives and outlining the methodology	LOUKMAN OMAR WAIS (L) ALOU COULIBALY	Fevrier-Mars 2025	The project concept has been developed, with its originality validated through a literature review. This process also helps define the next steps by clarifying the direction of the original idea.
2	- Identifying the methods to be used - Conducting relevant theoretical studies. - Proceeding to the design phase	ALOU COULIBALY (L) LOUKMAN OMAR WAIS	Fevrier-Mars 2025	The methods to be applied are supported by theoretical explanations. Continuing the design depends on the theoretical knowledge proficiency.
3	Completion of design calculations and drawings, along with the development of the system architecture and technical planning.	LOUKMAN OMAR WAIS (L) ALOU COULIBALY	April-May 2025	All necessary design calculations and technical schematics for the system have been finalized as part of the project development process.
4	- Developing the simulation model - Conducting simulation studies, analyzing the results, and documenting the findings in the Engineering Design Project.	ALOU COULIBALY (L) LOUKMAN OMAR WAIS	April-May 2025	A simulation model of the wind speed forecasting system was developed and implemented using MATLAB, where machine learning algorithms were applied and evaluated to assess predictive performance..
5	- Documenting the findings in the Engineering - Design Project report Including all steps: methodology, model setup, data processing, results, and discussion	LOUKMAN OMAR WAIS (L) ALOU COULIBALY	May 2025	It contributes by clearly documenting the entire project process, ensuring transparency, enabling replication, and supporting future improvements or research.

6	• Preparing slides and visual materials for the Engineering Design presentation • Presenting the project to the jury and responding to feedback • Participating in the Graduation Project Exhibition • Completing final exams	ALOU COULIBALY (L) LOUKMAN OMAR WAIS	May-June 2025	These activities contribute by effectively communicating the project outcomes, demonstrating presentation skills, engaging with professional feedback, and fulfilling academic requirements for graduation.
7	Final exams, Graduation Project Exhibition and presentations and completion of the project	LOUKMAN OMAR WAIS ALOU COULIBALY	June 2025	At the Graduation Project Exhibition, the system simulation was demonstrated in working condition and evaluated by the jury

Table 1.1. Work-Time Table

This Work-Time Table outlines the planned schedule for each phase of the project, including research, data preprocessing, model development, simulation, and reporting. It helps ensure timely progress, efficient task management, and successful project completion within the designated timeframe.

Table 1.2. Risk analysis and B plans

WP No	Risk Analysis	B Plans Solution
1	Incomplete or poor-quality historical wind speed data	Use publicly available datasets from reliable sources like or national meteorological institutes.
2	Missing or inconsistent meteorological parameters (humidity, pressure...)	Apply data imputation techniques or eliminate features with too many missing values.
3	MATLAB Deep Learning Toolbox limitations or bugs	Switch to Python (TensorFlow/Keras) if needed; ensure backups of code and models.
4	Difficulty in understanding LSTM implementation	Refer to tutorials and official documentation, or switch to other ML models (e.g., Random Forest, XGBoost).
5	Simulation results not aligning with expectations	Reevaluate preprocessing steps, consider adding more relevant features, or increase training data size.
6	Time constraints due to exams or other academic duties	Create a weekly timeline with buffer weeks; prioritize core tasks (data, model, simulation).
7	Hardware/software failure or data loss	Regularly back up data and code on cloud services (Google Drive, GitHub, etc.)..
8	Difficulty in interpreting and presenting results	Use clear visualizations (MATLAB plots), consult with academic supervisor for guidance.
9	Lack of access to real-time data (for future integration)	Simulate real-time conditions using time-segmented historical data.

1.8 Organization of Work Packages and Work Management

To ensure the project moved forward in a planned and timely way, all work packages were shared between the two team members. Each person was responsible for specific tasks within these packages, and all duties were completed in line with the timeline shown in Table 1.1. The first half of the work schedule was focused on research, theoretical studies, and design development, while the second half dealt with simulation, evaluation, and reporting. This clear organization of tasks helped us maintain steady progress and complete the project successfully and on time.

Table1.3. Distribution of work packages

Work Package Leader	Leading Work Packages	Number of Work Packages
LOUKMAN OMAR WAIS	1,5,11,26,39	36
ALOU COULIBALY	3,17,20,32,49	36

CHAPTER 2 : THEOROTICAL BACKGROUND

2. General Information

This project focuses on developing an AI-based system for forecasting wind speed to improve the efficiency of wind-powered energy systems. By using historical meteorological data and machine learning models, the system aims to predict wind speed accurately and support optimal microgrid management. When wind energy is insufficient, power is supplemented from the grid. This approach enhances the reliability and sustainability of wind energy integration.

2.1 Artificial intelligence

Artificial Intelligence (AI) is the field of computer science that focuses on creating systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, and decision-making. It encompasses various techniques, including Machine Learning (ML), where computers learn patterns from data to make predictions or automate processes. In this project, AI specifically through ML and neural networks is applied to forecast wind speed by analyzing past meteorological data and capturing complex patterns in the weather. The figure 3, illustrates the hierarchical relationship between Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL). Deep Learning is a subset of Machine Learning, which itself is a subset of the broader field of Artificial Intelligence.

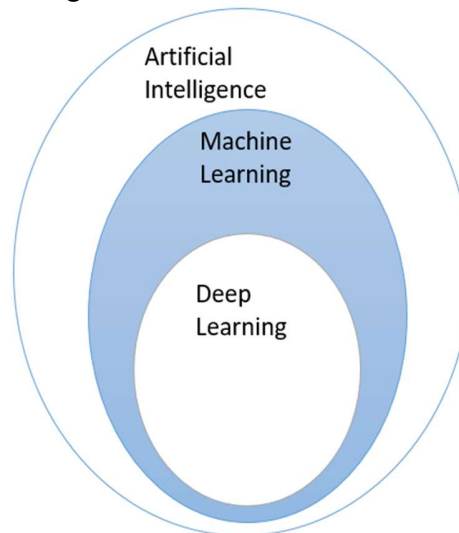


Figure 2.1. This diagram illustrates the nested relationship among Artificial Intelligence (AI), Machine Learning (ML), and Deep Learning (DL). Deep Learning is a specialized subset of

Machine Learning, which itself falls under the broader domain of AI. This hierarchy reflects how these technologies build upon each other to achieve intelligent behavior in computer systems.

2.2. Machine Learning

Machine Learning (ML) is a branch of Artificial Intelligence that enables computers to learn from data and improve their performance on tasks without being explicitly programmed. ML algorithms identify patterns in historical data and use these patterns to make predictions or decisions. In this project, Machine Learning is used to forecast short-term wind speed by analyzing meteorological inputs such as temperature, humidity, and pressure, allowing the model to learn and respond to complex environmental changes.

Machine Learning starts with two elements: an **algorithm** and a **data** set. The algorithm instructs the ML model on what to do (analyze images, detect patterns, and make predictions). The data set may or may not be **classified** or labeled to assist the algorithm. The algorithm then processes the data to produce an output.[8]

2.2.1. Types of Machine Learning

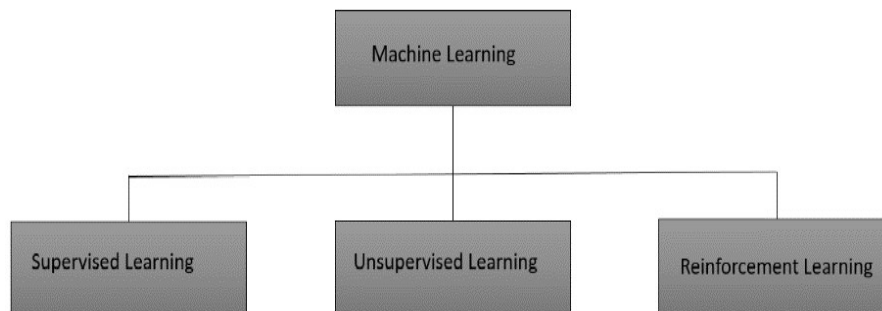


Figure 2.2.1. This diagram is showing the three main branches of machine learning: Supervised Learning, Unsupervised Learning, and Reinforcement Learning, all under the umbrella of Machine Learning. It provides a simple overview of how different learning paradigms are categorized.

2.2.2. Supervised learning

Supervised learning is a type of Machine Learning where a model is trained on a labeled dataset, meaning that each input is paired with a known output. The goal is for the model to learn the relationship between inputs and outputs so it can accurately predict outcomes for new, unseen data. In this project, supervised learning is used to train the LSTM model to forecast wind speed based on historical weather data, where features like temperature, humidity, and pressure are the inputs and the wind speed is the labeled output.[9]

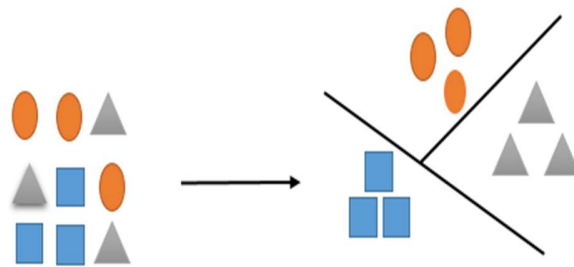


Figure 2.2.2. Overview of supervised learning. Input examples are categorized into a known set of classes.

a. Classification

It is a machine learning task where the model is trained on labeled data to predict a discrete class label. The goal is to categorize input data into predefined classes based on patterns learned from the training dataset. For example, cat or dog, positive or negative.

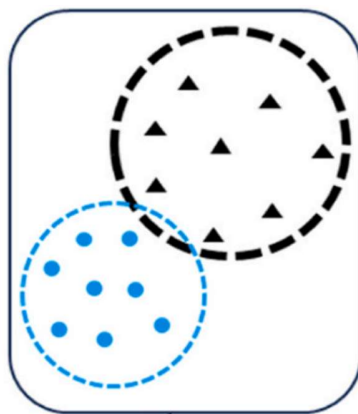


Figure 2.22.1. Where two distinct groups blue dots and black triangles are separated into clusters. The overlapping circles illustrate that some data points may lie near the boundary and need a model to classify them correctly.

b. Regression

It involves training a model on labeled data to predict a continuous numerical value (temperature or price). The model learns the relationship between input features and continuous output values to make accurate predictions on new data.

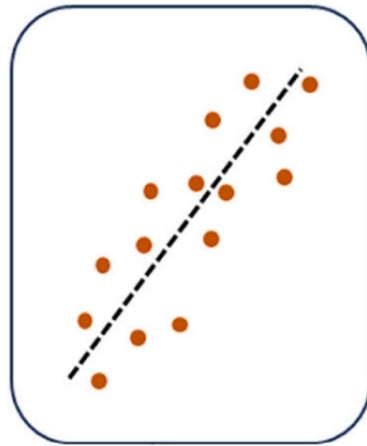


Figure 2.2.2.2 : The model learns the underlying relationship between the input features and the target variable, enabling it to make accurate predictions on new, unseen data. The regression line (or curve) in the diagram illustrates how the model fits the data, aiming to minimize the prediction error between actual and predicted values

2.2.3. Unsupervised learning

Unsupervised learning is a type of Machine Learning where the model is trained on data without labeled outputs. Instead of predicting specific outcomes, the model identifies patterns, structures, or groupings within the data. It is commonly used for tasks such as clustering, dimensionality reduction, and anomaly detection. In the context of meteorological data, unsupervised learning can help discover hidden relationships or natural groupings in weather patterns without prior knowledge of the target variable. Figure 4, It can analyze animal data and group the animals by their traits and behavior. These groups could correspond to different species, making it possible to organize the animals without pre-existing labels.[10]

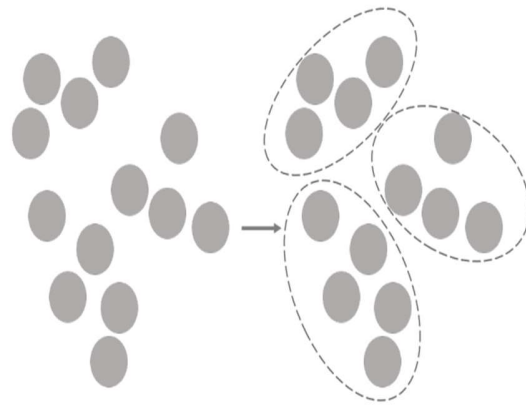


Figure 2.2.3. Overview of unsupervised learning. Input samples are grouped into clusters based on the underlying patterns.

a. Clustering

It is a machine learning technique where data points are grouped into clusters based on their similarities, without predefined labels. This method helps uncover hidden patterns or structures in data, such as grouping customers with similar purchasing behaviors for targeted marketing.

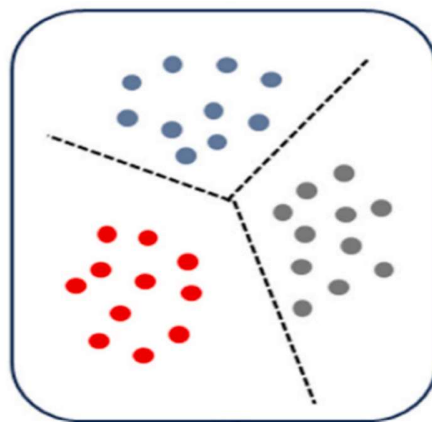


Figure 2.2.3.1 : Overview of unsupervised learning. Input samples are grouped into cluster based on the underlying patterns.

b. Association

Association rule learning is an unsupervised method used to discover interesting relationships, patterns, or associations among variables in large datasets. Commonly applied in market basket analysis, it identifies rules like "if a customer buys bread and butter, they are likely to also buy milk," aiding in decision-making processes.

2.2.4 Reinforcement learning

It is a machine learning paradigm where an agent learns to make decisions by interacting with an environment, aiming to maximize cumulative rewards through trial and error. Unlike supervised learning, RL does not require labeled input/output pairs; instead, it relies on feedback from the environment to guide learning.

2.3 Machine learning working principle

Machine learning algorithms are essential tools for identifying and utilizing patterns and relationships within datasets. The learning process begins by training the algorithm on a dataset composed of numerous data samples. During this phase, the algorithm examines these samples to detect consistent patterns and connections among the data[11]. Training involves supplying the algorithm with input data and the corresponding target outputs. Through repeated adjustments to its internal parameters, the algorithm aims to reduce the difference between its predictions and the actual values in the dataset. This ongoing refinement process helps the model to more accurately represent the intricate relationships present in the data.

After the model has been thoroughly trained, it can apply what it has learned to new, unseen data. This capability, known as generalization, is vital as it enables the algorithm to perform tasks such as prediction or classification on data it has not encountered before. For example, if trained on a dataset containing labeled images, the model could then classify new images based on their visual content.

For this project, the machine learning model is trained using a dataset that includes various variables related to wind energy, solar power, and energy pricing. These variables might include wind speed and direction, solar irradiance, and other environmental or economic factors. By analyzing these inputs, the model identifies trends and dependencies that enhance its ability to make reliable predictions in real-world energy scenarios. To summarize, machine learning algorithms extract meaningful patterns from historical data, enabling them to apply these insights to future data for accurate forecasting or decision-making.

CHAPTER 3 : DATA COLLECTION AND PREPROCESSING

3.1. General Information

In this In this project, an AI-based wind energy forecasting system is developed to estimate wind speed availability, enhancing the planning and operation of wind-powered systems. The methodology begins with the collection and preprocessing of historical meteorological data, including variables such as wind speed at 10 m and 100 m, terrestrial radiation, wind direction at 10 m, surface pressure, temperature, humidity, and wind gusts at 10 m. After cleaning, normalizing, and selecting relevant features, the data is divided into training and testing sets. To accurately forecast wind speed, various machine learning models are explored and evaluated based on performance metrics such as RMSE, MAE, and R-squared. Among these, the Support Vector Machine (SVM) model demonstrates the highest accuracy. MATLAB is used as the development environment due to its robust tools for model training, validation, and performance analysis. By capturing complex relationships and dependencies within the data, the implemented AI models significantly improve the reliability of short-term wind speed forecasts, ultimately supporting efficient microgrid energy management and renewable integration.

3.2. Data collection

In this study, data collection serves as a foundational step for building and evaluating predictive models. Accurate and high-resolution meteorological data are essential for understanding weather conditions and training models capable of identifying patterns in atmospheric behavior. To ensure data quality and relevance, the information used in this thesis was sourced from reliable, real-time observation stations, capturing key environmental parameters at minute-level intervals.

The dataset [12] used in this study was obtained from the Open Meteo database. It was collected from Istanbul (Turkiye) observation station .The observation period spans from January 1st, 2020, at 00:00:00 to January 1st, 2025, at 23:00:00, with data collected every hour during this time frame.

The dataset contains input variables or features and output variable, input variables are the variables which are used to predict the output variable which is the goal of the study.

The input recorded for each observation (instance) include :

- Observation date and time
- Wind direction at 100 m
- Wind speed at 10 m and 100 m
- Temperature 2m
- Relative humidity 2m
- Terrestrial radiation
- wind gusts 10m

The output of those elements is :

- The wind speed
- Observation date and Time : refers to the specific timeframe during which data is collected or observed.
- Wind Speed at 100m and at 10 m : it is a measure of the rate at which air is moving in the atmosphere. These features represent how fast the air is moving at heights of 100 meters and 10 meters above the ground, respectively. Wind speed is typically measured in meters per second (m/s) and is a crucial parameter in evaluating wind energy potential and understanding atmospheric dynamics.
- Wind direction at 100 m : wind direction is a measure of the compass direction from which the wind originates. This feature refers to the direction from which the wind is blowing at a height of 100 meters above the ground. It is usually measured in degrees, where 0° indicates a northerly wind, 90° an easterly wind, 180° a southerly wind, and 270° a westerly wind. Understanding wind direction is essential for applications such as wind energy generation, weather forecasting, and environmental monitoring.
- Temperature at 2 m,: is a measure of the thermal energy or heat present in the atmosphere. This feature specifically represents the air temperature measured at 2 meters above ground level. It is typically recorded in degrees Celsius (°C) and is essential for analyzing weather patterns and environmental conditions.
- Relative humidity : is the ratio of the current amount of water vapor in the air to the maximum amount the air can hold at that temperature, expressed as a percentage. This feature indicates

the relative humidity measured at a height of 2 meters above the ground. It is crucial for understanding atmospheric moisture, weather patterns, and human comfort.

- Terrestrial radiation : refers to the longwave radiation emitted by the Earth's surface as it loses heat absorbed from the sun. This feature measures the energy emitted from the ground into the atmosphere, usually in watts per square meter (W/m^2). It is an important factor in the Earth's energy balance and surface temperature regulation.
- Surface pressure : it is the atmospheric pressure measured at the Earth's surface, usually in pascals (Pa). It plays a key role in wind forecasting, as wind is driven by differences in pressure. Including surface pressure as a feature in machine learning models helps improve wind speed predictions by capturing important weather patterns and pressure gradients that influence wind behavior.
- Wind gust at 10 m : wind gusts are sudden, brief increases in wind speed, often caused by turbulence or weather events. This feature represents the maximum wind gusts recorded at a height of 10 meters above the ground. Gusts are typically measured in meters per second (m/s) and are important for assessing extreme weather conditions and their potential impact.

Table 3.2

Date & Time (UTC)	Temp (°C)	RH (%)	Wind Speed 10m (m/s)	Surface Pressure (Pa)	Terrestrial Radiation (W/m ²)	Wind Dir 100m (°)	Wind Speed 100m (m/s)	Wind Gusts 10m (m/s)
2024-07-20 00:00:00	25.5455	87.66	13.58	1005.41	0.0	35.64	23.48	22.68
2024-07-20 01:00:00	25.4955	91.43	10.5	1005.51	0.0	15.07	19.39	16.92
2024-07-20 02:00:00	25.6955	88.47	13.91	1005.32	0.0	23.43	23.54	21.6
2024-07-20 03:00:00	25.6455	87.4	15.63	1005.42	0.86	30.83	25.99	25.2
2024-07-20 04:00:00	26.6455	80.91	19.08	1005.53	140.19	37.61	27.72	31.68
2024-07-20 05:00:00	27.9455	73.62	18.67	1005.44	383.36	40.33	25.03	33.84
2024-07-20 06:00:00	29.3455	64.85	20.15	1005.26	621.23	42.17	25.74	37.8
2024-07-20 07:00:00	30.5455	57.82	21.76	1005.38	837.6	40.19	27.33	41.76
2024-07-20 08:00:00	31.4455	53.1	23.05	1005.39	1017.7	39.99	29.13	43.92
2024-07-20 09:00:00	31.7955	51.89	23.11	1005.39	1149.26	37.88	28.73	44.64

Table 3.2 presents hourly meteorological datasets before preprocessing collected over 9-hour period on 20-07-2024, from the Open Meteo weather station in Istanbul.

3.3. Data preprocessing

Before model training, the dataset needs to be prepared so that it is suitable for the model before feeding it into a machine learning. It's one of the most important steps in the AI workflow because poor-quality input data often leads to poor model performance.

This involves handling missing data, transforming data, scaling the data, and splitting the data.

In this section, I will explain each preprocessing step individually, highlighting its importance and how it contributes to improving model performance.

Each of these steps will be explained in detail and at the end of the explanation, all the preprocessing techniques will be combined into a single Python code block that loads the dataset

and applies the necessary cleaning and transformation operations. This ensures that the data is fully prepared for the model training phase.

3.3.1 Data Normalization

In machine learning, the training process involves iterative optimization based on feature values. If the input features are on vastly different scales, this can negatively impact the learning **process**, especially for algorithms that rely on distance metrics or gradient descent (such as neural networks, logistic regression, or support vector machines).

To mitigate this, we apply data normalization to rescale features to a common range, ensuring that no single feature disproportionately influences the model due to its numerical magnitude.

In this study, the optimization algorithm (gradient descent) updates weights based on the magnitude of feature values. If one feature has values in the range of thousands (pressure), and another in decimal points (radiation), the optimization process may converge slowly or erratically, prioritizing larger-scale features.

Normalization ensures that :

- All features contribute equally to the cost function.
- The training converges faster and more reliably.
- The model is less sensitive to the units of measurement or the original distribution of data.

In this project, we employ Min-Max Normalization, which transforms each feature to lie within the range 0 and 1. This technique is particularly well-suited for models that are sensitive to feature scales, such as neural networks.

$$X_{\text{normalized}} = \frac{X - X_{\min}}{X_{\max} - X_{\min}} \quad \text{Equation (1)}$$

Where:

- X_{norme} is an original feature value
- $X_{\min}$ and $X_{\max}$ are the minimum and maximum values in that feature column.

3.3.2. Data Cleaning and Transformation (Handling missing values)

As mentioned in Table 3.2 wind direction is measured in degrees, which follow a circular pattern. However, using degrees directly in a machine learning model can cause problems and can make it hard for the model to understand certain characteristics of the data. For example, if we have two wind directions of 350° and 10°, the model might interpret them as very far from each other, despite them being very close in reality [13].

To avoid this, one common approach is to transform the wind direction into their corresponding cosine and sine values to capture the cyclical patterns. As an example, a wind direction of 180° would be represented with its cosine value (-1) and sine value (0) [14].

This was done by first converting the wind direction values from degrees to radians using Equation:

$$\text{Radians} = \text{degrees} \times \pi / 180 \quad \text{Equation (2)}$$

After applying the conversion formula to transform wind direction from degrees to radians, the sine and cosine values of the resulting radians were calculated. These values were then used to create two new features: wind_dir_sin and wind_dir_cos. Finally, the original wind direction column was removed from the dataset, as its information was now fully represented by the new features.

3.3.3 Reducing Skew

Reducing skew is the process of transforming data to make its distribution more symmetric, often to improve the performance and reliability of statistical analyses and machine learning models.

In this study during the initial exploratory data analysis of the meteorological datasets collected from Izmir, Ankara, and Denizli, it was observed that the features Wind Speed (m/s) exhibited a notable positive skew across all three locations. To normalize these distributions and reduce the impact of skewness, a logarithmic transformation was applied.

The log transformation is defined by the following equation :

$$X' = \log(X + 1) \quad \text{(Equation 3)}$$

This formulation ensures that zero and near-zero values are handled without introducing undefined results in the transformation. In the case of the Denizli dataset, additional features including Temperature (°C), Humidity (%), and Wind Direction (°) demonstrated significant negative skewness. To handle the angular nature of Wind Direction (°), the feature was decomposed into its sine and cosine components using:

$$\text{Sin}(\theta) = \sin(\pi/180 \times \text{Wind Direction } (^\circ)) \quad \text{Equation (4)}$$

$$\text{Cos}(\theta) = \cos(\pi/180 \times \text{Wind Direction } (^\circ)) \quad \text{Equation (5)}$$

wind direction exhibited a positive skew. Consequently, the log transformation (Equation 2) was applied to the sine values to correct the skew. The features which showed negative skewness in the dataset, a log transformation was deemed unsuitable, as it would further distort the distribution. Instead, a square transformation was applied to these variables, as described by the While this transformation allowed directional data to be used effectively in modeling, the sine-transformed following equation:

$$\mathbf{X}' = \mathbf{X}^2 \quad \text{Equation (6)}$$

This transformation helps to correct negative skewness by increasing the spread of larger values more than smaller ones. These preprocessing steps were crucial for improving the performance and reliability of the predictive models by ensuring that key meteorological inputs followed more symmetrical distributions.

3.3.4. Data Scaling

Scaling the data involves transforming the values of all features to have the same scale, which helps prevent the model from being biased toward features with larger values [15]. Scaling the data can improve the performance of machine learning. One common method for scaling the data is min-max scaling [15], which involves scaling the features to a fixed range, for instance between 0 and 1. This is done by subtracting the minimum value and dividing by the difference between the maximum and minimum values [16].

The formula for min-max scaling is shown in Equation [6] :

$$X_{\text{scaled}} = \left(\frac{X - X_{\min}}{X_{\max} - X_{\min}} \right) \times (b - a) + a \quad \text{Equation (7)}$$

Where:

- X_{scaled} is the scaled value,
- X is the original value,
- X_{min} is the minimum value of the feature in the dataset,
- X_{max} is the maximum value of the feature in the dataset.

Multiplying the result with 2 and subtracting 1 scales the values to the range $[-1, 1]$. When scaling the data, it is important to avoid data leakage, which means that information from the validation and test sets is leaked into the training set. To prevent data leakage, the minimum and maximum value for each input variable were calculated using the training set only. By doing so, the training of the model is not influenced by any information from the validation or test sets

3.3.5. Creating create Lagged Features

Creating lagged features is a method used to transform time series data into a supervised learning format by shifting each variable's values backward in time. This transformation is essential before training machine learning models, as it enables the model to learn from historical patterns in the data.

In time series forecasting, especially for multivariate datasets, this technique allows the model to predict future values based on past observations. Since the aim of this project is to predict wind speed using various meteorological features,

the dataset must be structured accordingly. Each feature including the target variable requires its own set of lagged features, which are generated by shifting the original data forward by a specified number of time steps.

This process effectively transforms the dataset into a supervised learning problem, enabling the model to capture temporal dependencies and improve forecasting accuracy.

For example, to create the lagged feature for $t-1$, the original series is shifted forward one time step. For $t-2$, the original series is shifted forward two time steps, and so on. By creating these lagged features, we can provide the model with information about past values of each feature, which can help improve its predictions.

$$A(t-1) \quad B(t-1) \quad A(t) \quad B(t)$$

0	10	1	20
1	20	2	30
2	30	3	40
3	40	4	50

There is no rule for the optimal number of lagged features needed for a specific number of forecasting steps, which can also vary depending on the specific characteristics of the time series.

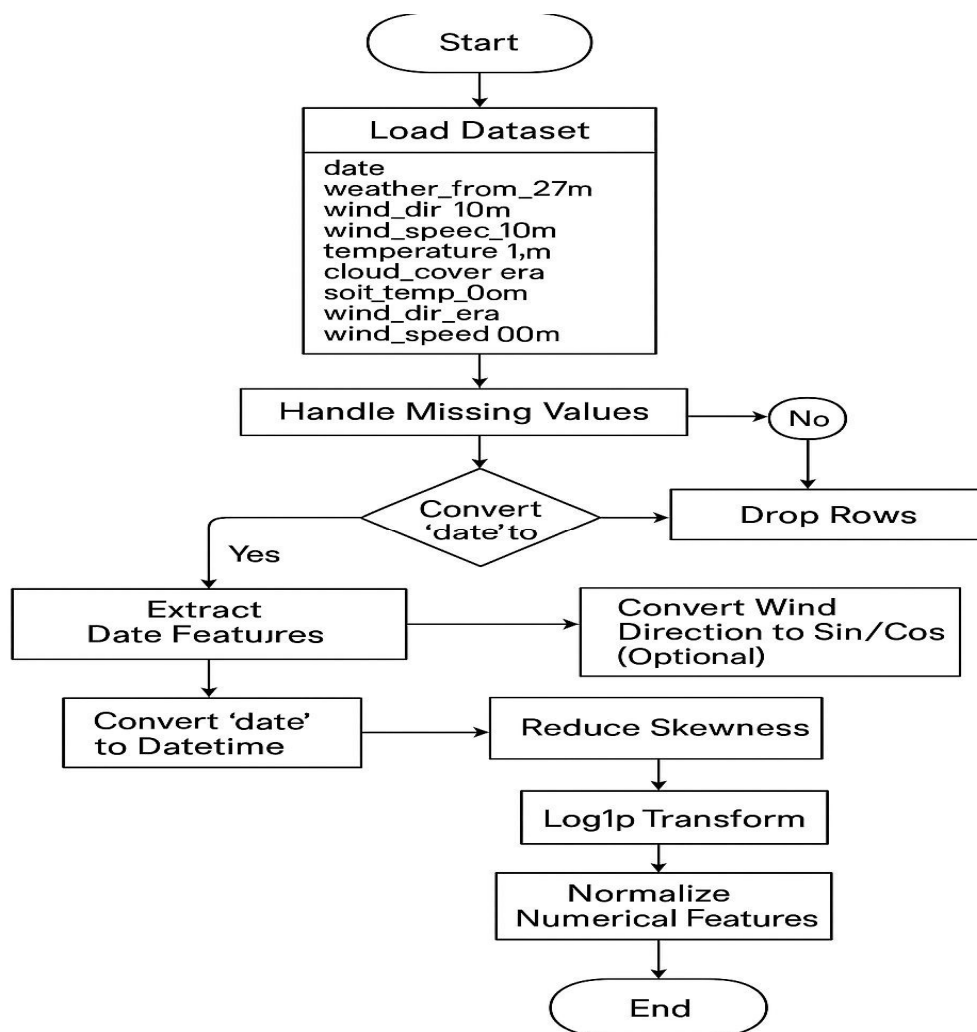


Figure 3.3 : Data Preprocessing Flowchart, this flowchart outlines the sequential steps taken to clean and transform the dataset, including handling missing values, feature extraction, skewness correction, and normalization.

CHAPTER 4: MODEL SELECTION AND TRAINING

4.1. Model selection

We plan to evaluate six distinct models to gain a thorough insight into the dataset's properties and the predictive capabilities of each approach. An AI model refers to a system that utilizes algorithms to analyze data, identify patterns, and autonomously make predictions or decisions without human input. Accurate wind speed forecasting is pivotal for optimizing energy production in wind farms and enhancing meteorological predictions. Various machine learning models offer distinct advantages depending on the nature of the data and the specific forecasting requirements. This section provides an overview of several prominent models considered for this study.

4.1.1. Decision tree model

Decision Trees are a non-parametric supervised learning method used for classification and regression the goal is to create a model that predicts the value of the target variable by learning simple decision rule inferred from the data features[17]. A tree can be seen as piecewise constant approximation. The decision tree is a non-parametric supervised learning algorithm renowned for its interpretability and capability to model complex patterns in data. Structurally, a decision tree resembles a flowchart, where internal nodes represent "conditions" (also referred to as splits or tests), branches indicate outcomes of these conditions, and leaf nodes correspond to predictions or final outcomes. Each decision path from the root to a leaf encodes a series of logical rules, making the model highly transparent and easily understandable. This hierarchical structure enables decision trees to recursively partition the input space, facilitating both classification and regression tasks.

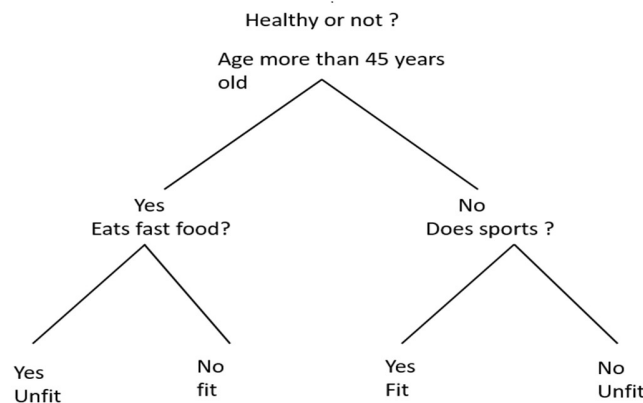


Figure 4.1.1: It illustrates a classification decision tree that predicts an individual's health status (Fit or unfit) based on a series of binary decisions.

It begins by evaluating whether the person is older than 45 years. If so, it further checks if the individual consumes fast food to determine the final classification. If the person is 45 or younger, the model assesses whether they engage in sports. Each internal node represents a decision rule, and each leaf node assigns a class label.

The tree demonstrates how simple, interpretable rules can be used to classify data based on health-related attributes.

4.1.2. Regression tree model

Linear regression is a foundational supervised learning algorithm designed to model the linear relationship between a dependent variable and one or more independent variables[18]. It achieves this by fitting a linear equation to the observed data. When only a single independent variable is involved, the approach is termed Simple Linear Regression; with multiple predictors, it becomes Multiple Linear Regression. Similarly, linear regression can be univariate when there is a single target variable or multivariate when multiple outcomes are modeled simultaneously. The principal objective of linear regression is to determine the optimal line of best fit that minimizes the difference (error) between the predicted and actual values.

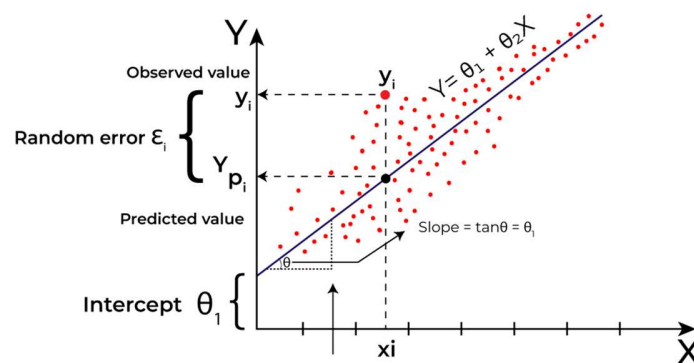


Figure 4.1.2 : It presented above, the variable Y represents the dependent or target variable, while X denotes the independent variable, also referred to as the predictor of Y . There are various forms of functions that can be applied in regression analysis, among which the linear function is the most fundamental. In this context, X may represent a single feature or a set of multiple features that characterize the underlying problem. Linear regression aims to estimate the value of the dependent variable (Y) based on the input of the independent variable (X), which justifies the term *Linear Regression*.

The resulting regression line, known as the line of best fit, represents the model that minimizes the error between predicted and actual values. The mathematical form of the linear regression model is expressed as follows:

$$Y = \theta_1 + \theta_2 X \quad \text{Equation (8)}$$

In the equation:

- Y is dependent variable
- X is independent variable
- θ_1 is intercept or slope
- θ_2 is the regression coefficient

4.1.3. Support vector machine model

Support Vector Machine (SVM) is a robust and versatile supervised learning algorithm that effectively models both linear and non-linear relationships between input features and target variables. It is widely applied in both classification and regression contexts, offering strong generalization capabilities across a range of problems[19]. The central objective of SVM is to identify an optimal decision boundary referred to as a hyper plane that distinctly separates data points belonging to different classes within an n-dimensional feature space. The algorithm selects specific data points, known as support vectors, which lie closest to the hyper plane. These support vectors are pivotal in defining the position and orientation of the hyper plane, and they give the algorithm its name. A key concept in SVM is the margin, which is defined as the distance between the hyper plane and the nearest data points from either class. The ideal hyper plane is the one that maximizes this margin, thereby improving the classifier's ability to generalize to unseen data. For example, in a binary classification scenario with two classes Say, Orange and blue the algorithm evaluates multiple possible separating lines and selects the one that achieves the greatest distance from the closest points of both classes. This maximization of the margin leads to improved classification robustness.

SVM can be categorized into two main types:

- **Linear SVM:** This variant is suitable for linearly separable data, where a single straight line (or hyper plane in higher dimensions) can successfully distinguish between the classes.

- **Non-Linear SVM:** When the data is not linearly separable, when a straight line is insufficient to separate the classes, advanced methods such as kernel functions are employed. These kernel tricks transform the input data into higher-dimensional spaces where a linear separator may then be found.

Through these mechanisms, SVM demonstrates strong flexibility and precision in various machine learning tasks, particularly in high-dimensional and complex datasets.

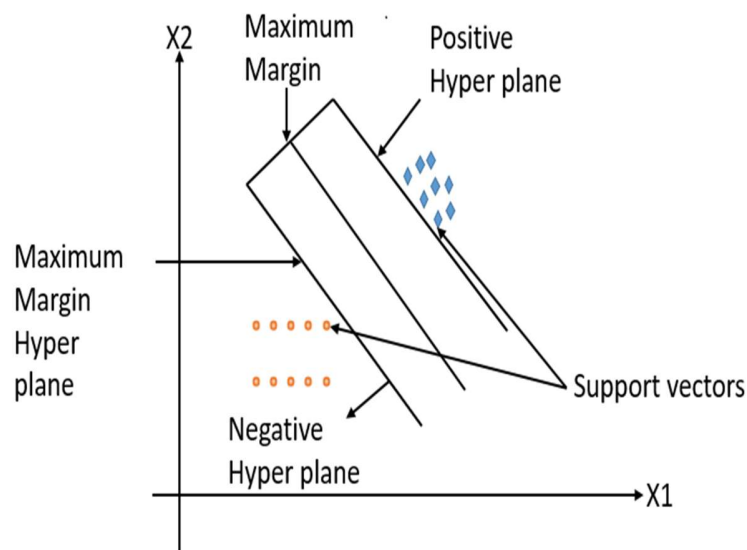


Figure 4.1.3: An SVM (Support Vector Machine) figure is provided to illustrate the fundamental operating principle of this model. Consider a dataset comprising two distinct classes (orange and blue). The objective is to classify a new data point as belonging to either class. Although multiple decision boundaries could potentially separate the classes, the critical question lies in identifying the most optimal one. The optimal decision boundary, referred to as the hyper plane, is the one that maximizes the margin the distance between the hyper plane and the nearest data points from each class. This is the central goal of the SVM algorithm. SVM evaluates various possible hyper planes and selects the one with the greatest separation from the closest data points of each class, thereby ensuring maximum generalization capability. The algorithm is categorized into two main types: Linear SVM, which is applicable when the data is linearly separable, allowing for a single straight line (in two dimensions) to distinctly divide the classes; and Non-linear SVM, which is employed when the data cannot be separated by a straight line. In such cases, more advanced techniques,

such as kernel methods, are utilized to transform the input space into a higher-dimensional space where a linear separation becomes possible.

4.1.4 Ensemble Methods model :

Ensemble learning refers to a class of machine learning techniques that combine predictions from multiple base models to improve overall accuracy and robustness[20]. The fundamental principle is that a group of diverse models can often outperform a single predictive model by compensating for each other's biases and errors. Ensemble approaches enhance predictive performance and generalization, especially in uncertain or noisy environments.

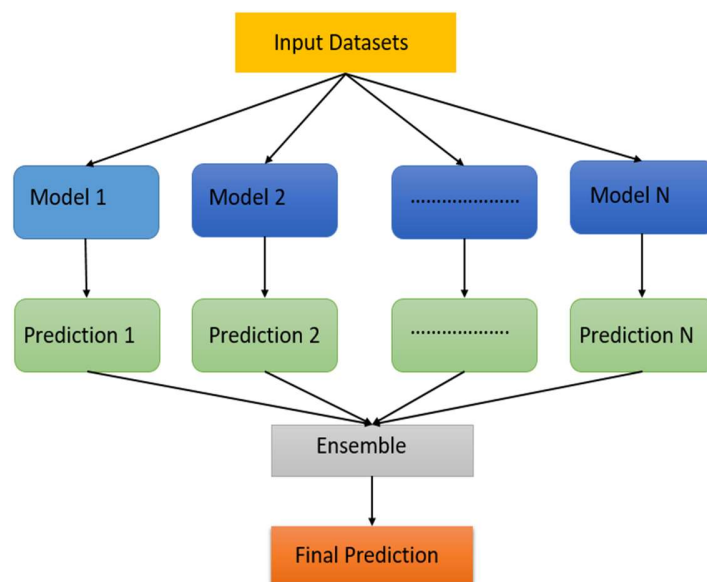


Figure 4.1.4: The stacking approach, a type of ensemble learning technique, functions by enabling a meta-learning algorithm to integrate the predictions generated by multiple base-level models of similar or diverse types.

Ensemble method has three main types :

i. Bagging

Bagging is a commonly used ensemble technique, especially in classification and regression tasks. It works by constructing multiple models typically decision trees trained on different random subsets of the training data. These subsets are generated using bootstrapping, a sampling method that selects data points with replacement, thereby ensuring variability among training sets. This

process reduces model variance, thus improving stability and mitigating over fitting a common challenge in high-variance models. The aggregation step in bagging involves synthesizing the predictions from all the base models, often through majority voting in classification tasks or averaging in regression problems. By aggregating these predictions, the ensemble produces a more balanced and accurate result than any single model could achieve individually.

ii. Boosting

Boosting is another ensemble approach that focuses on incrementally improving the model's performance by sequentially training models. Each successive model attempts to correct the prediction errors made by its predecessor[23]. The final output is a weighted sum of all the models' predictions, with greater emphasis placed on models that perform better. Several boosting algorithms exist, including Adaptive Boosting (Ada Boost), gradient boosting, and Extreme Gradient Boosting (XG Boost).

- **Ada Boost:** employs simple base learners known as decision stumps trees with a single split and assigns higher weights to incorrectly classified instances, ensuring subsequent learners focus more on these challenging cases.
- **Gradient Boosting:** builds new learners to predict the residuals (errors) of previous learners, optimizing the model using gradient descent techniques.
- **XG Boost:** further enhances the gradient boosting process by incorporating regularization and parallel computing, making it both faster and more accurate in practice, though often computationally intensive.

a. Stacking

Stacking is a sophisticated ensemble learning strategy that combines the predictions of multiple base learners using a meta-learner, which attempts to identify the most accurate blend of predictions. Unlike bagging and boosting, which combine similar types of learners, stacking allows for a diverse mix of model architectures. The base models generate individual predictions, which are then used as inputs to train the meta-model. Stacking has been effectively utilized in tasks such as regression, classification, density estimation, and even in more complex learning environments. By learning how to optimally integrate various models, stacking often surpasses the performance of any single constituent model.

a. Maximum voting :

Predominantly applied in classification problems, this method involves each model casting a "vote" for a predicted class. The class that receives the most votes becomes the final prediction.

b. Averaging :

Common in regression tasks, this method calculates the mean of the predictions from all models. It can also be adapted for classification by averaging predicted probabilities.

c. Weighted average :

An enhancement of simple averaging, this technique assigns different weights to models based on their individual performance or importance. The final prediction is computed as a weighted sum of the individual predictions, thereby giving more influence to the more reliable models.

4.1.5. Neural network model :

Neural networks are computational frameworks inspired by the architecture of the human brain. They comprise layers of interconnected artificial neurons (nodes), each capable of processing input data and passing the result to subsequent layers. These connections are weighted, and through training, the network learns to adjust these weights to minimize prediction error[21]. Neural networks are adept at modeling non-linear relationships and identifying complex patterns in data, making them valuable for tasks like image recognition, natural language processing, and time-series forecasting.

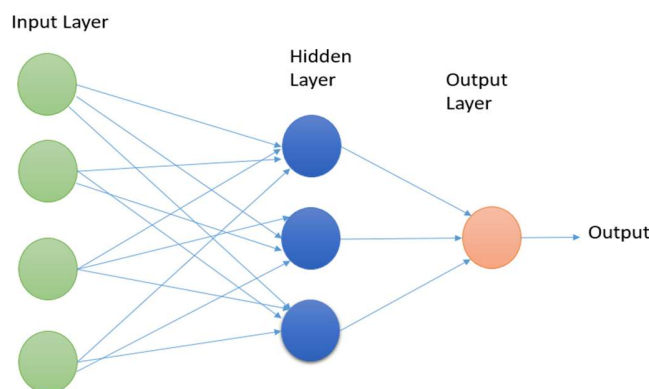


Figure 4.1.5: It illustrates the operational mechanism of a neural network. In this architecture, each neuron in the first hidden layer receives input from all nodes in the preceding input layer,

establishing full connectivity. Likewise, every neuron in the hidden layer is connected to each input feature through weighted links. This pattern continues across successive layers; neurons in the second hidden layer treat the outputs of the first hidden layer as their inputs and maintain similar full connectivity. This sequential computation, where data flows from the input layer through hidden layers to the output layer, is known as forward propagation. During this phase, the neural network processes input data to generate predictions. Subsequently, the predicted output is compared against the actual target values. This comparison facilitates the calculation of the prediction error, which is essential for the learning process in neural networks.

4.1.6. Efficient Learning Models

Efficient learning models are optimized variants of standard algorithms, such as linear regression, tailored to handle large-scale datasets and high-dimensional feature spaces with improved computational efficiency. A prominent approach within this paradigm is reinforcement learning, wherein algorithms learn optimal strategies through interaction with their environment via a reward-based mechanism. The agent iteratively explores actions, receiving feedback in the form of rewards (for desired outcomes) or penalties (for incorrect decisions), gradually improving performance over time without explicit programming. This paradigm is especially valuable in applications requiring real-time adaptation and decision-making under uncertainty[22].

4.2 Model training

In this section we will train models mentioned above and compare the experimental result yielded by those models.

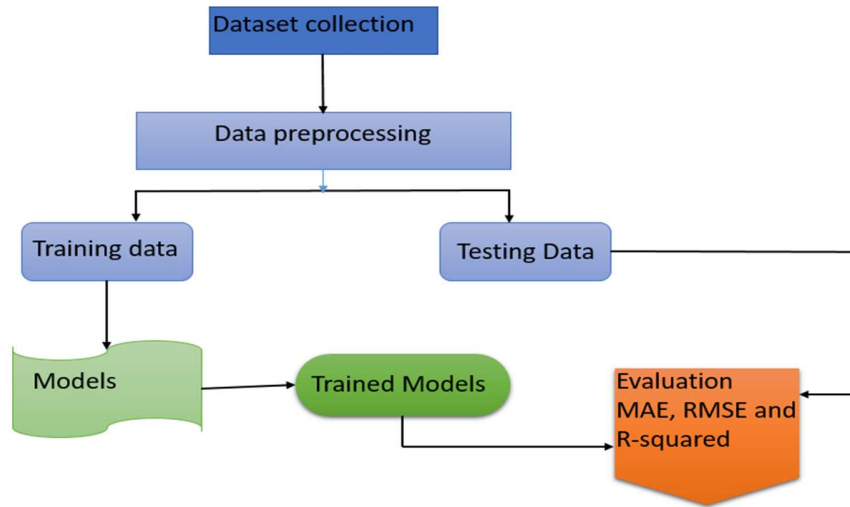


Figure 4.2: Shows the whole process structure of the procedure

Initially, wind or solar energy datasets are collected and subjected to preprocessing, which includes data cleaning and normalization to ensure consistency and reliability. The processed data is then divided into two subsets: a training set, which is used to develop and fit the predictive models, and a testing set, which serves to evaluate the performance of the trained models. The aforementioned models are trained on the training data, and their effectiveness is assessed using appropriate error metrics. These metrics help in comparing the models and identifying the one with the highest predictive accuracy. Training a diverse set of six regression models allows for a comprehensive analysis of the dataset's behavior and its forecasting capabilities. Each model yields different perspectives and performance indicators, supporting a more informed selection of the most appropriate model for deployment, taking into account the specific objectives and limitations of the application.

4.2.1. Feature engineering :

Feature engineering represents a vital phase in the machine learning workflow, wherein raw data is systematically transformed into informative variables that improve the predictive capability of models. Specifically, in the context of renewable energy forecasting, this process entails identifying, constructing, and refining features that effectively capture underlying trends and correlations within the dataset. Before initiating the training of models, feature engineering was carried out to isolate and modify pertinent input variables. This involved the extraction of critical attributes from the dataset, followed by the application of suitable transformations aimed at

optimizing model performance. Transforming raw inputs into a format conducive to modeling is essential for ensuring algorithmic efficiency and accuracy. Common transformation techniques include feature scaling, normalization, and the encoding of categorical data. Feature scaling ensures that all variables contribute equally by placing them on a uniform scale, while normalization adjusts feature values to fall within a standard range, often between 0 and 1. Among various normalization techniques, two widely used approaches are:

1. **Min-Max Scaling:** This method linearly rescales features to a fixed range, typically (0, 1).
2. **Z-Score Normalization:** This technique standardizes features based on their mean and standard deviation. In this study, Min-Max normalization was employed using the following formula:

$$X' = \frac{X - X_{\min}}{X_{\max} - X_{\min}}$$

Equation (9)

- X: original value,
- X min: minimum value of the data feature
- X max: maximum value of the data feature

This normalization ensures that all input features lie within the same scale, thus enhancing the learning efficiency and convergence of the models.

4.3 Performance evaluation matrix

The performance of the trained models was assessed using established evaluation metrics, namely Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). These metrics were selected to quantitatively measure the predictive accuracy and robustness of the models in estimating renewable energy parameters.

4.3.1 Mean absolute error MAE

The Mean Absolute Error (MAE) is a widely used evaluation metric for quantifying the average magnitude of errors between predicted and actual values in regression analysis. It is calculated as the mean of the absolute differences between predicted outputs and their corresponding true

values, thereby disregarding the direction of the errors. This metric provides a direct and interpretable measure of prediction accuracy, expressed in the same units as the target variable. For example, an MAE of 2.5 indicates that, on average, the model's predictions deviate from the actual values by 2.5 units. MAE is particularly valuable due to its simplicity and robustness; unlike metrics such as Root Mean Squared Error (RMSE), it is less sensitive to large deviations or outliers, as it does not involve squaring the errors. The formal definition of MAE is expressed as follows:

Where:

$$x' = \frac{x - x_{\min}}{x_{\max} - x_{\min}} \quad \text{Equation (10)}$$

- X: original value,
- X min: minimum value of the data feature
- X max: maximum value of the data feature

A lower MAE value reflects a more accurate model, as it indicates that the predictions are, on average, closer to the actual outcomes. Conversely, a higher MAE signifies larger average errors and thus lower predictive accuracy.

4.3.2 Root mean square error

Root Mean Squared Error (RMSE), also referred to as Root Mean Square Deviation, is a commonly employed metric for assessing the performance of regression models. Similar in purpose to Mean Absolute Error (MAE), RMSE differs by assigning greater importance to larger errors due to the squaring of differences between predicted and actual values. Specifically, RMSE is computed as the square root of the mean of the squared deviations, which accentuates the impact of substantial prediction errors on the overall performance metric. This characteristic makes RMSE particularly useful for evaluating the dispersion and variability in model predictions, as it reflects how concentrated the residuals are around the actual values. A lower RMSE value signifies improved model accuracy, while a higher RMSE indicates greater prediction error. However, due to the squaring component, RMSE is more sensitive to outliers compared to MAE.

The RMSE is calculated using the following formula:

$$\mathbf{MAE} = \frac{\sum_{i=1}^n |y_i - x_i|}{n}$$

Equation (11)

Where:

- Y_i represents the predicted value of the target variable for the i th observation.
- X_i represents the actual value of the target variable for the i th observation.
- N represents the total number of observations in the dataset.

This metric provides a clear indication of the average magnitude of error between predicted and true values, making it a valuable tool for regression model evaluation.

The equation facilitates the quantitative assessment of model performance by highlighting discrepancies in prediction accuracy.

4.3.3 R-squared

The R-squared statistic (denoted as R^2), also known as the coefficient of determination, is a key metric used to assess the explanatory power of regression models. It quantifies the proportion of variance in the dependent (response) variable that is accounted for by the independent (predictor) variables. The value of R^2 ranges between 0 and 1, where a value of 1 indicates that the model perfectly explains the observed variability in the target variable, while a value of 0 implies that the model fails to capture any of the variability. Mathematically, R^2 is determined by comparing the residual sum of squares (RSS), which measures the unexplained variation, to the total sum of squares (TSS), which captures the overall variation in the dependent variable. The formula for R^2 is given as[24]:

$$\mathbf{R^2 = 1 - \frac{Rss}{Tss}}$$

Equation (12)

Where:

- The Residual Sum of Squares, representing the sum of squared differences between actual values and predicted values.

$$\mathbf{Rss} = \Sigma(\mathbf{x_i} - \mathbf{y_i})^2 \quad \text{Equation (14)}$$

Xi = actual value

Yi = predicted value

The Total Sum of Squares, representing the total variance, with being the mean of the observed values.

$$\mathbf{Tss} = \Sigma(\mathbf{x_i} - \bar{\mathbf{y}})^2 \quad \text{Equation (15)}$$

Xi = actual value

Yi = mean of the observed value

A higher R^2 value indicates a greater proportion of the variance in the dependent variable is explained by the model, thereby reflecting a better model fit. However, it is important to note that R^2 can be artificially inflated by simply adding more predictors to the model, regardless of their relevance.

To counteract this issue, the Adjusted R-squared metric is often used, as it incorporates a penalty for the inclusion of non-significant variables, offering a more reliable measure of model performance when comparing models with different numbers of predictors.

Table 5.2

Wind Speed, v (m/s)	Mechanic Torque, $T_m(\text{Nm})$	Generator Speed, $w_G(\text{rad/s})$	Line-to-line voltage at generator, $U(\text{V})$	Line Current, $I_H (\text{A})$	Load power $P_{YÜK}=UI$ (W)
6	-205.6	0.2366	84.88	0.392	33.27
8	-365.5	0.2499	96.22	0.4444	42.76
10	-571.1	0.267	114.3	0.528	60.36
12	-822.3	0.2878	139.2	0.6431	89.54
14	-1119	0.3122	176	0.8131	143.10

Table 5.2 : The power output of a wind energy system is highly sensitive to variations in wind speed. As illustrated in Table 1, an increase in wind speed from 6 m/s to 14 m/s results in a significant rise in electrical power output, from 33.27 W to 143.10 W. This sharp increase is attributed to the cubic relationship between wind speed and wind power potential, expressed. Thus, even small increments in wind speed lead to disproportionately larger increases in power generation. The data clearly demonstrate that higher wind speeds enable the turbine to harness greater kinetic energy from the wind, thereby enhancing the electrical power delivered to the load. This relationship underscores the importance of wind speed as a critical factor in the efficient operation and energy yield of wind power systems.

5.3. Models training

This section presents the training results of various regression models using the MATLAB Regression Learner tool. The models were trained on the dataset provided above to predict a target variable.

5.3.1. True vs predicted plot:

A true versus predicted plot is a type of scatter plot used to evaluate the performance of a regression model. In this plot, the actual values are shown on the x-axis, while the predicted values are plotted on the y-axis. When the model makes accurate predictions, the data points align closely along a diagonal line with a slope of 1, indicating a perfect match between predicted and actual values.

Ideally, if the predictions are perfect, the points will lie along a straight line with a slope of 1. [25]

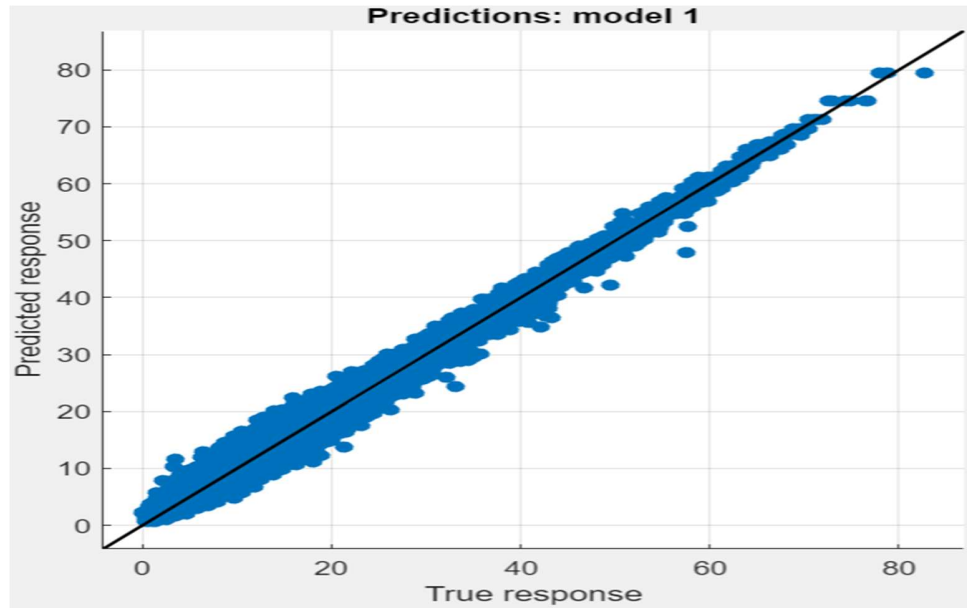


Figure 5.3.1 True response VS Predicted response

Figure illustrates a plot comparing the true response values with the predicted responses generated by the model. As observed from the figure, the model demonstrates good performance, as most data points closely align along the diagonal slope line, indicating a strong agreement between actual and predicted values

5.3.2. Residual plot

A residual plot is a useful tool in regression analysis for assessing model performance. It displays the residuals the differences between the observed values and the model's predictions against either the predicted values or the input variables. For a well-fitted model, the residuals should appear randomly distributed around the horizontal line at $y = 0$. Any visible patterns or trends in the plot may suggest that the model has missed some important aspects of the data's structure.[26]

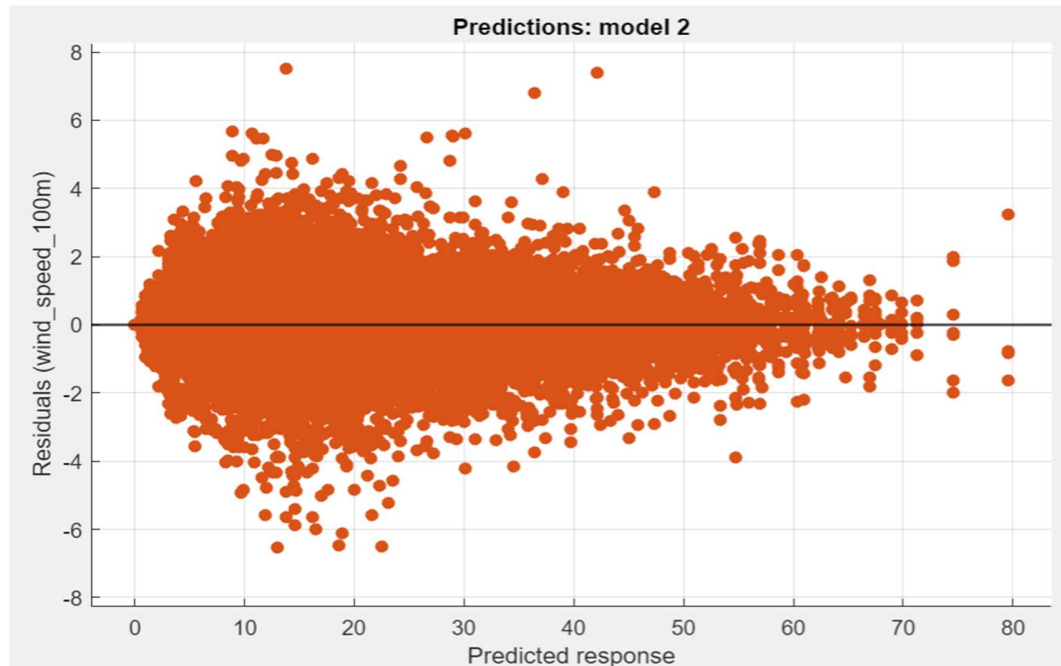


Figure 5.3.2 : Residual vs predicted response

Figure: this residual plot shows that Model 2's predictions are generally unbiased, as residuals are centered around zero. However, the spread of residuals is wider at lower predicted wind speeds and narrows at higher speeds, indicating heteroscedasticity. This means the model is less accurate for low wind speeds and more consistent for high wind speeds.

The above figure shows residual plot of a model which is predicting well

5.4. Simulation Models

In this part, various machine learning models were developed and tested to forecast wind energy parameters. The simulation phase involved training six different supervised learning algorithms Decision Tree, Linear Regression, Support Vector Machine (SVM), Neural Networks, Ensemble Models (Bagging, Boosting, Stacking), and Efficient Learning Models on preprocessed meteorological datasets.

Each model was evaluated using performance metrics including Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and the Coefficient of Determination (R^2). This multi-model approach provided a comprehensive performance comparison, enabling selection of the most accurate and reliable model for real-world wind energy forecasting applications.

5.4.1 Optimizable tree model

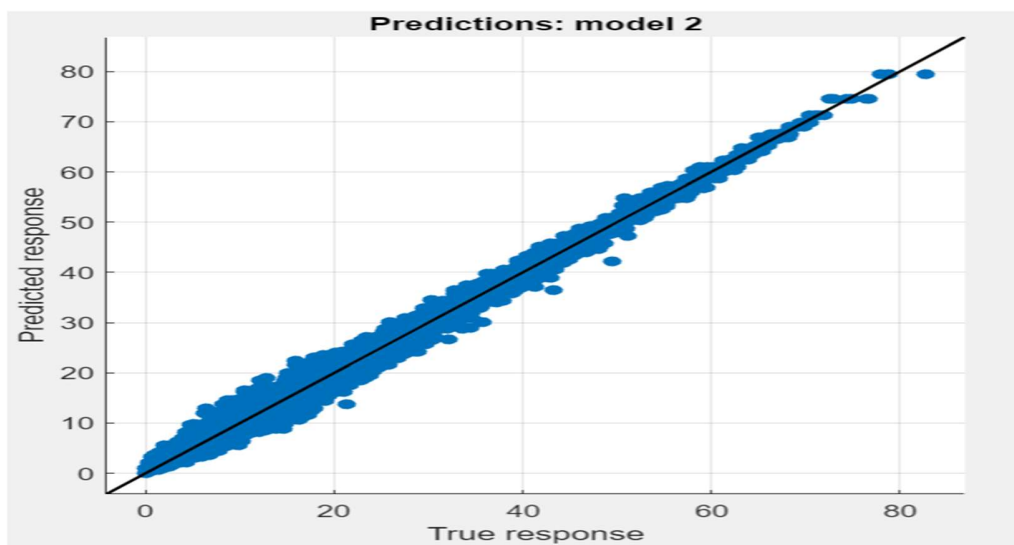


Figure 5.4.1.1 :Optimizable tree model true vs predicted response

The figure illustrates the performance of the Fine Tree regression model by visualizing its predicted values against the actual data. Identified as the best performing model in our study, the Fine Tree model demonstrates a strong alignment of data points along the ideal slope line, indicating that its predictions closely correspond to the true values. This close alignment reflects the model's high predictive accuracy and reliability. Overall, the results suggest that the Fine Tree model effectively captures the underlying patterns in the data across most cases.

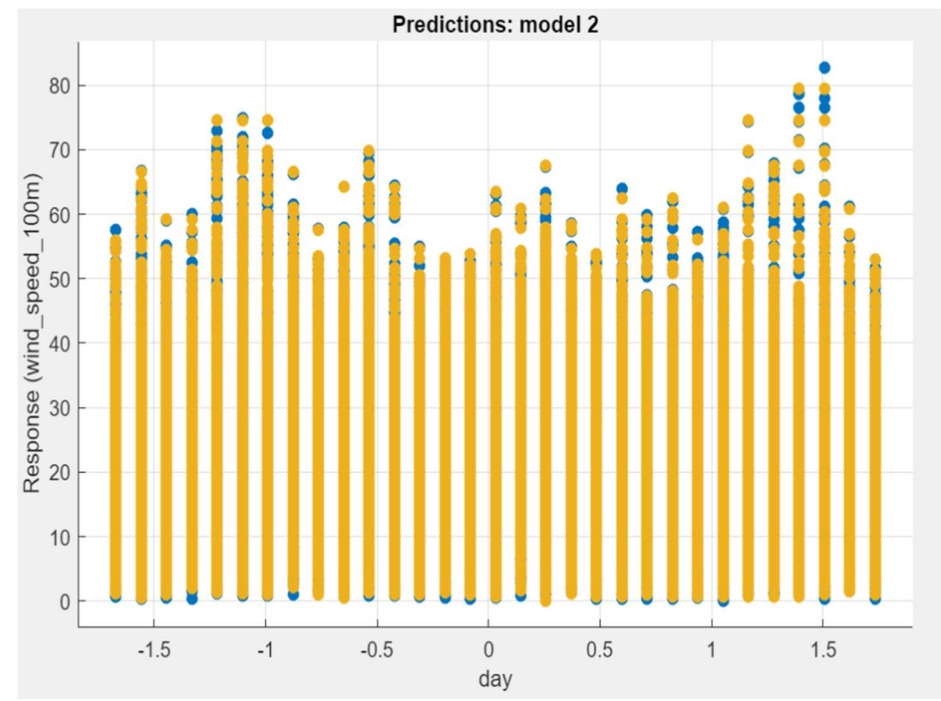
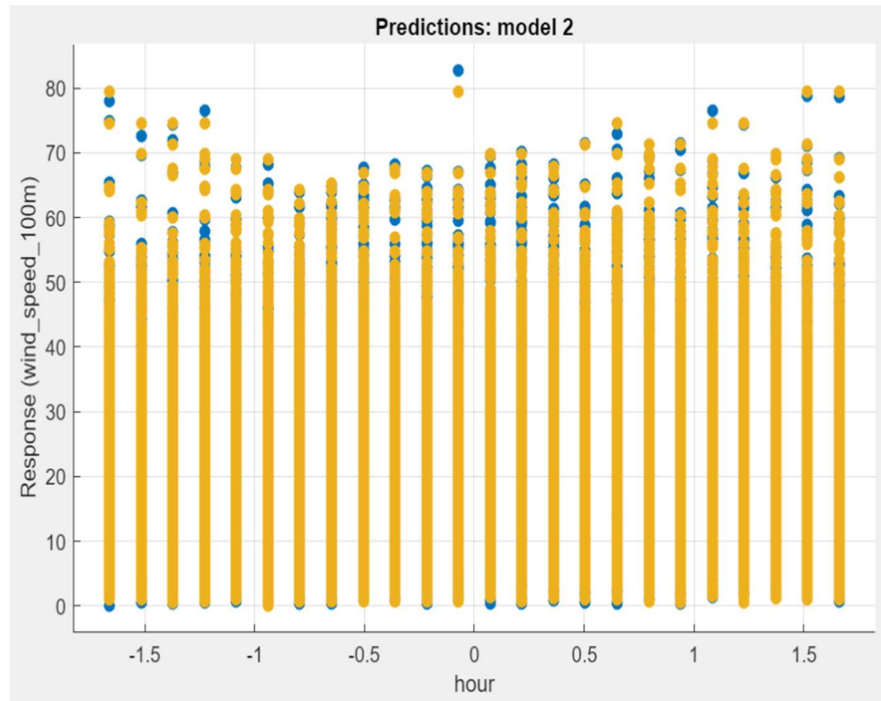


Figure 5.4.1.2 : Wind speed 100 m Response vs wind speed hourly and daily

The figures above present the prediction performance of the optimizable tree model developed for forecasting wind speed at a height of 100 meters. The left graph illustrates wind speed predictions across different hours of the day, while the right graph shows the model's performance across different days. In both visualizations, the orange bars represent the model's predicted values, and the colored dots correspond to the actual observed data.

The optimizable tree model successfully captures both hourly (diurnal) and daily fluctuations in wind speed. The close match between the predicted and actual values indicates that the model is effective in learning temporal wind speed patterns.

5.4.2 Linear Regression

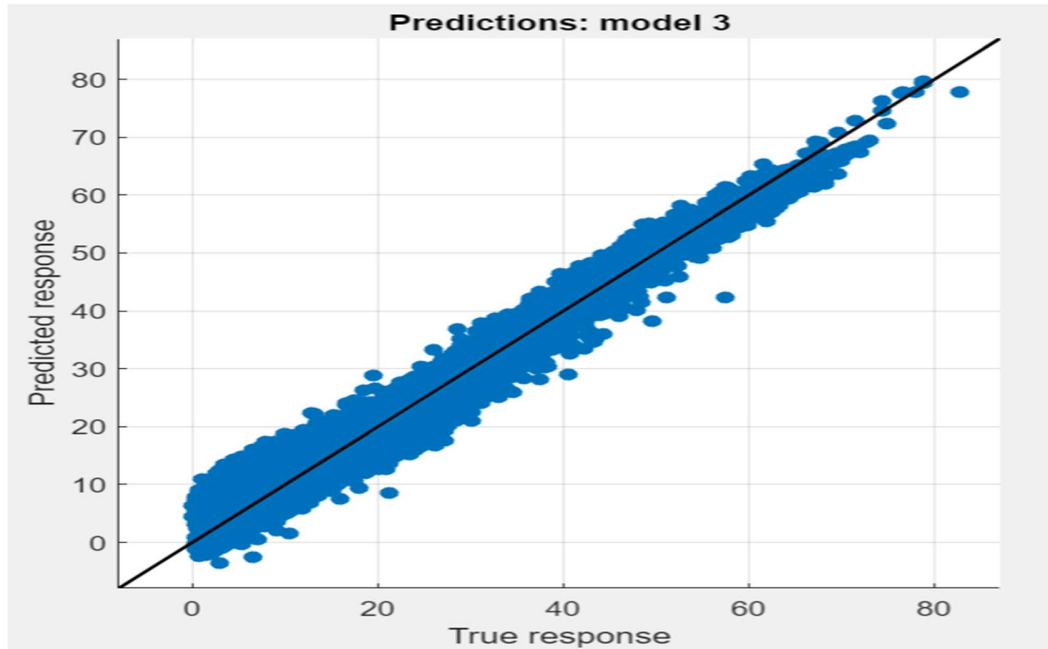


Figure 5.4.2.1 : Linear regression model true vs predicted

Figure 3.9 illustrates the performance of the Linear Regression model. The spread of data points shows noticeable deviations from the ideal diagonal line, highlighting differences between the predicted and actual values. Similar to the Fine Tree model, the Linear Regression model displays some inconsistency in its predictions, indicating challenges in fully capturing the data's underlying structure. This observation emphasizes the need for further model refinement to improve prediction accuracy and reduce errors in future application

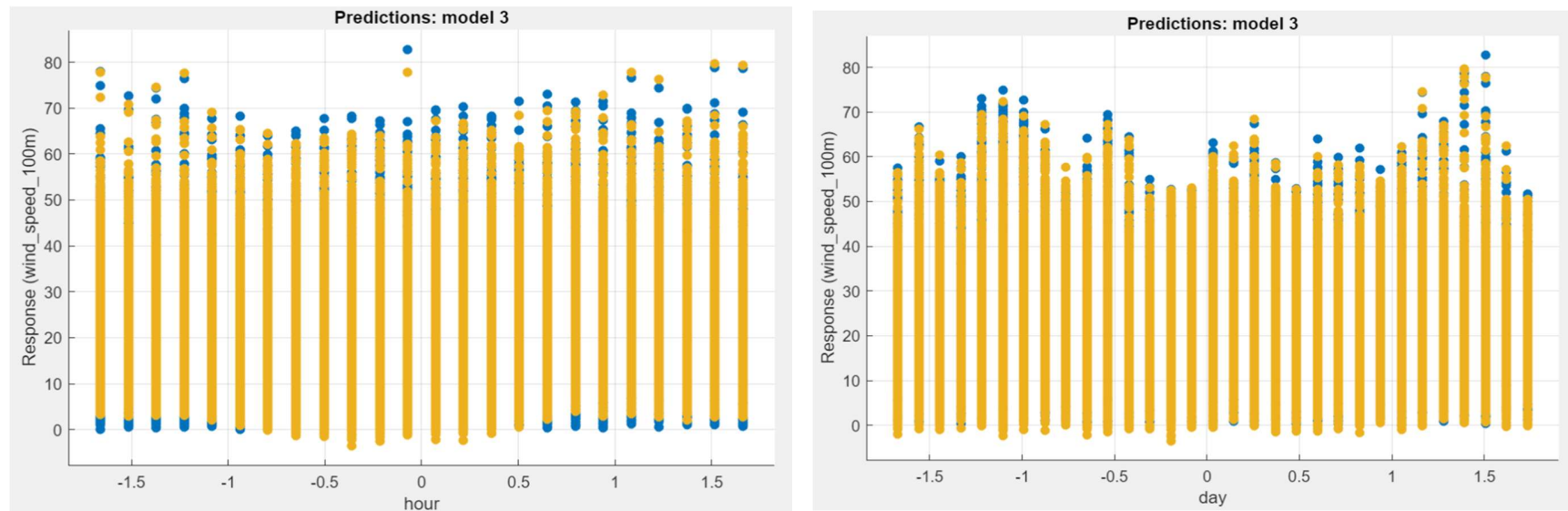


Figure 5.4.2.2 : Wind speed 100 m Response vs wind speed hourly and daily

The figures above show the performance of the linear regression model (Model 3) in forecasting wind speed at a height of 100 meters. The left graph represents predictions versus actual values across different hours, while the right graph shows the same comparison over different days. In both graphs, the yellow bars indicate the predicted wind speeds, and the blue dots represent the actual observed values. Despite the simplicity of linear regression, the model manages to capture the general trends in wind speed over time. However, compared to more complex models, its predictions show less sensitivity to fluctuations and may underestimate or overestimate wind speed during more variable conditions. This highlights both the strengths and limitations of using a linear approach for time-dependent wind forecasting. Overall, linear regression provides a baseline model that performs adequately for general trend estimation, but may require more advanced methods for capturing nonlinear patterns in wind behavior.

5.4.3 Efficient Linear

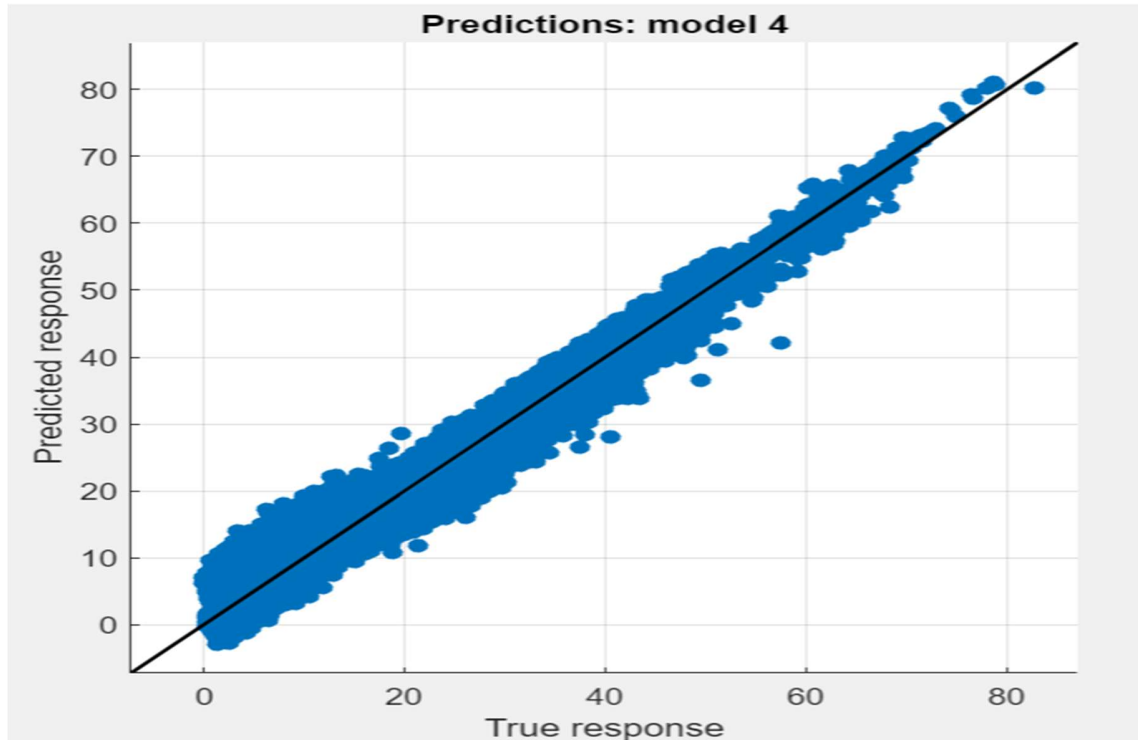


Figure 5.4.3.1 : Efficient learning model True vs predicted responses

The performance of the Efficient Learning model (Model 4) is depicted in this figure. The concentration of data points around the vertical line corresponding to the predicted values indicates that the model consistently predicts values near a certain range. However, this tight clustering also reveals limited variation in the model's outputs, which may reflect weak predictive performance and an inability to accurately capture the full range of the actual data.

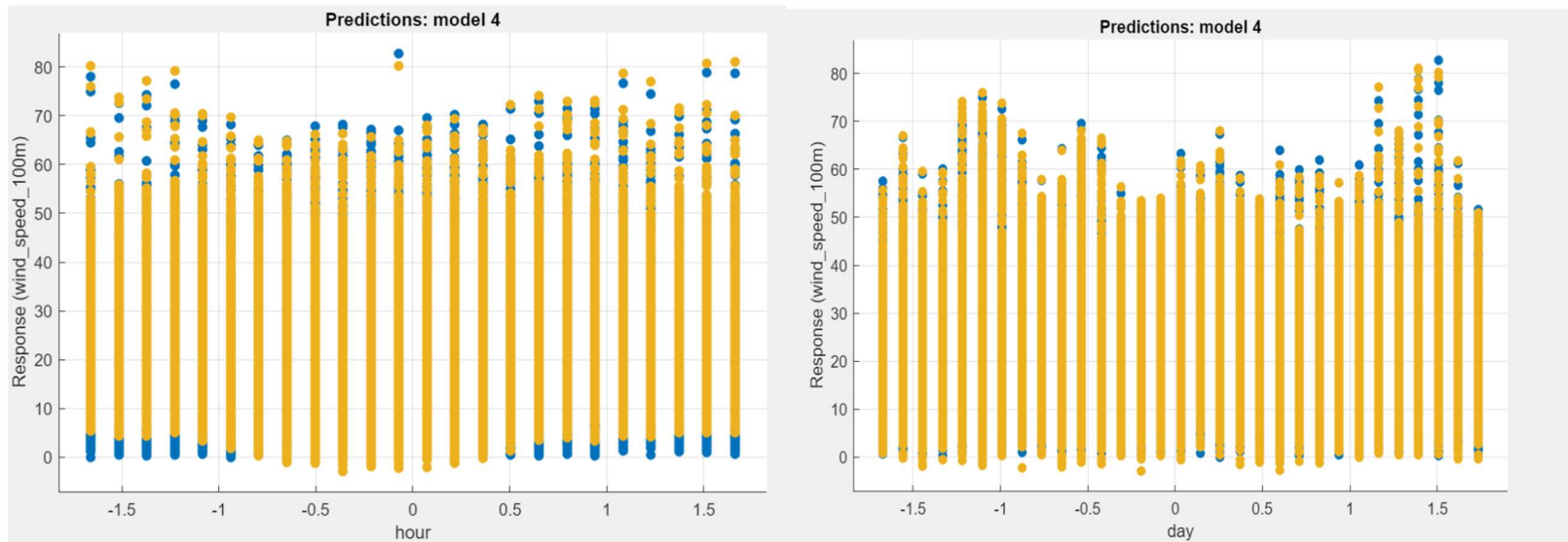


Figure 5.4.3.2: Wind speed 100 m Response vs wind speed hourly and daily

The graphs display the predicted vs. actual wind speed at 100 m using Model 4. The left graph shows performance by hour, while the right shows performance by day. Although the predictions roughly follow the trend of the true values, especially by hour, the model shows larger deviations across days.

Among all the models tested, Model 4 performs the worst, with a relatively high prediction error. Its evaluation metrics confirm this RMSE: 2.4193, MAE: 1.8765 and R^2 is 0.96

Despite the R^2 value appearing strong, the higher RMSE and MAE values compared to other models indicate less accurate predictions. These results suggest the model fails to capture daily patterns effectively, and improvements are needed for better generalization over time.

5.4.4 Neural Network

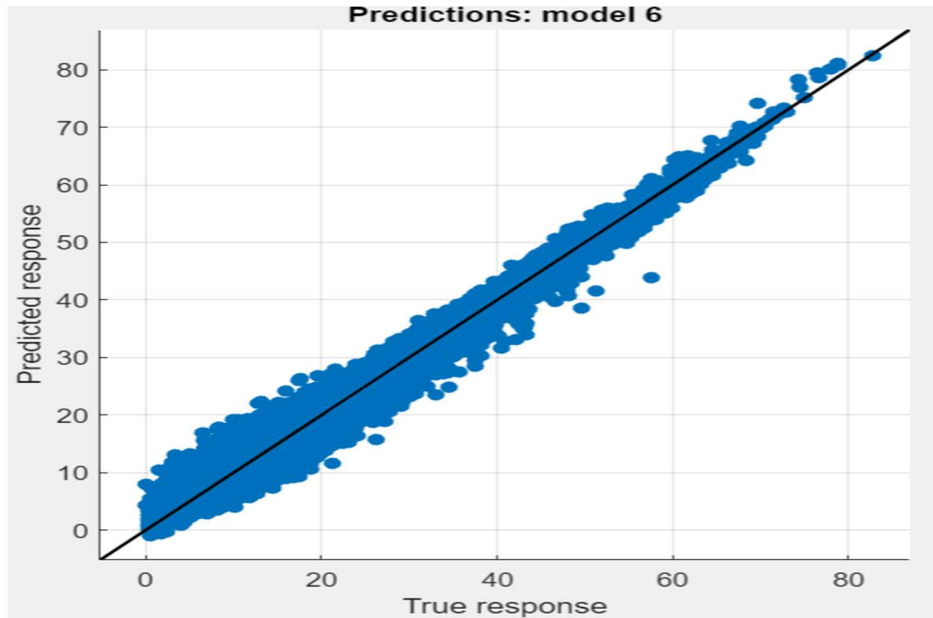


Figure 5.4.4.1 : NN model True vs predicted responses

The figure illustrates the performance of the neural network model, showing a dense concentration of data points around the vertical line representing predicted values. This clustering indicates that the model tends to generate predictions within a relatively narrow range. While this consistency can be a sign of stability, the limited spread of predictions suggests that the model may not fully capture the variability inherent in the actual data, potentially leading to suboptimal generalization.

Nevertheless, it is important to note that this model represents our second-best prediction in terms of overall performance. Despite its narrower prediction range, it achieves a lower error compared to other models, with the exception of the top-performing one. This balance of error performance and prediction consistency makes it a strong candidate in our evaluation.

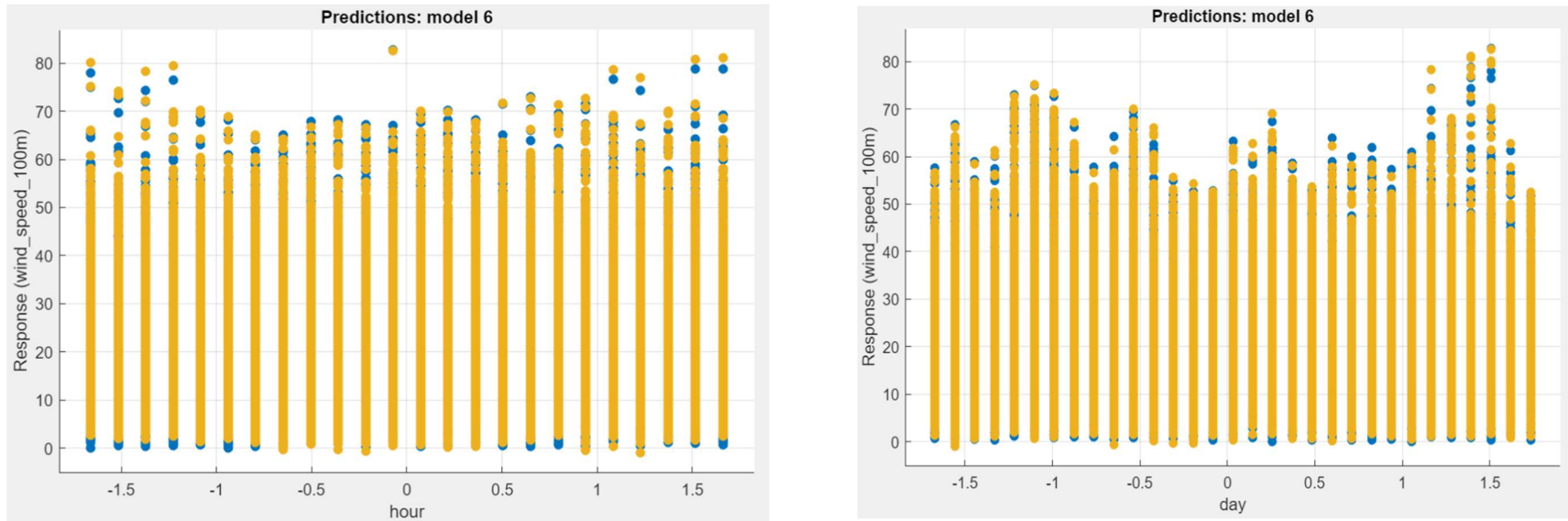


Figure 5.4.4.2 : Wind speed 100 m Response vs wind speed hourly and daily

The graphs present the performance of Model 6 (Neural Network). The model demonstrates stable and consistent predictions across both hour and day. Predicted values (blue) closely follow the true values (Yellow), especially across the hourly scale.

However, the predictions are often clustered within a narrow range, suggesting limited sensitivity to extreme variations in wind speed. This can lead to underestimation in certain high-variance periods, especially by day.

Despite this limitation, the model offers strong performance and generalization, achieving the second-best error metrics among all models. It strikes a good balance between prediction stability and accuracy, making it a reliable candidate in our evaluation.

5.4.5 Ensemble methods

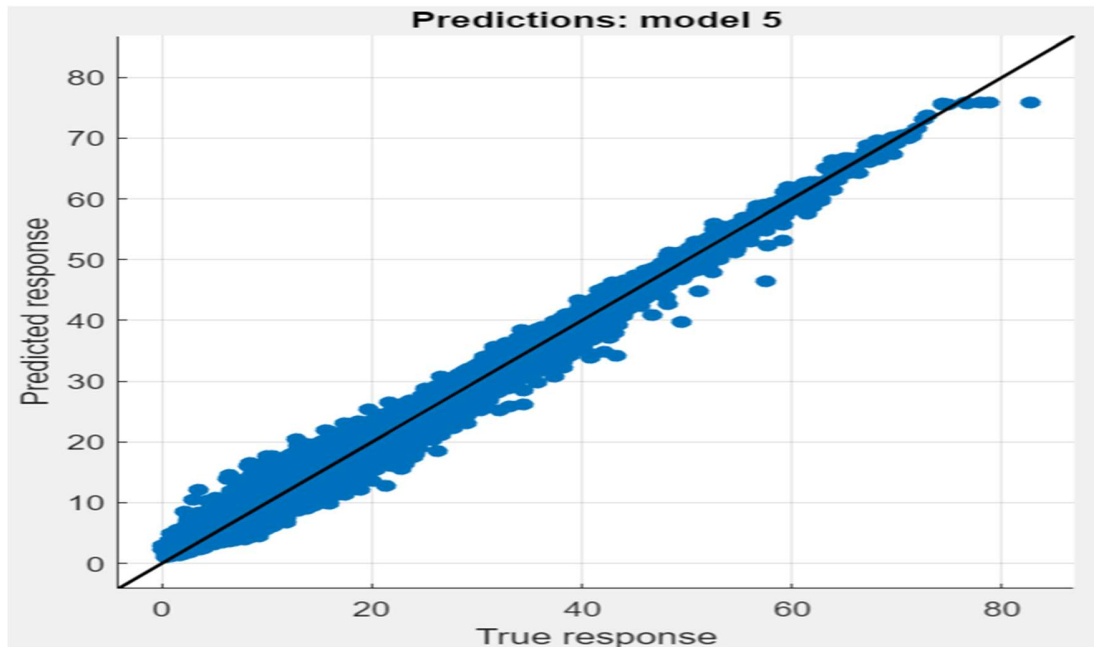


Figure 5.4.5.1 : Ensemble methods True vs predicted responses

The ensemble methods model demonstrated strong predictive capabilities, as evidenced by both graphical and statistical evaluation metrics. The scatter plot of true vs. predicted responses (Figure: Ensemble methods True vs predicted responses) shows that the model's predictions align closely with the actual values. The majority of data points lie along the 45-degree diagonal line, which represents perfect predictions. This visual pattern confirms that the ensemble model effectively captures the underlying trends in the data with minimal deviation. Quantitatively, the model achieved an R-squared (R^2) value of 0.98, indicating that it explains 98% of the variance in the true responses. This high R^2 suggests a strong goodness of fit. The Root Mean Squared Error (RMSE) of **1.6593** and Mean Absolute Error (MAE) of **1.1942** further reflect the model's low prediction error rates and consistent accuracy. Compared to other models evaluated, the ensemble methods model performed significantly better than linear and efficient linear models, and was only marginally outperformed by the optimizable tree model and neural network model. Nonetheless, its balance between accuracy, stability, and robustness makes it a reliable choice for regression tasks.

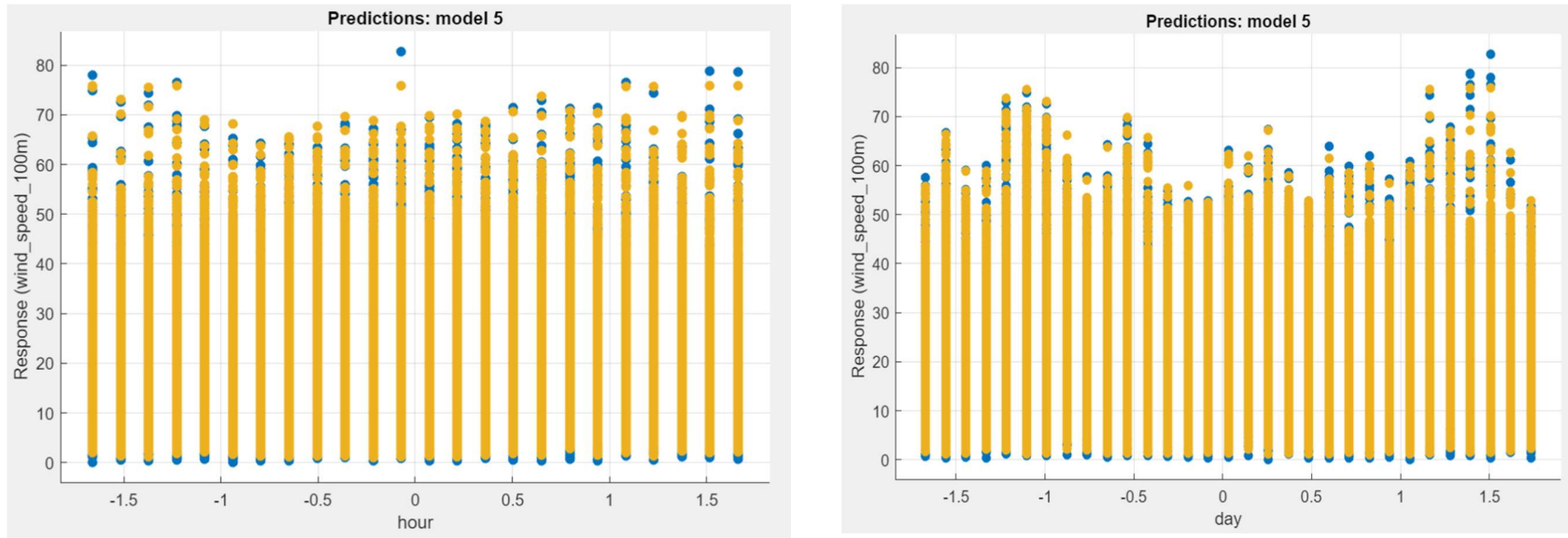


Figure 5.4.5.2 Wind speed 100 m Response vs wind speed hourly and daily

The figure shows Model 7's predictions of wind speed at 100 meters across different hours and days. The model closely follows the actual wind speed patterns, with predicted values (orange) aligning well with true values (blue). It captures daily and hourly variations effectively, demonstrating good accuracy and generalization over time. This indicates the model's strong potential for reliable wind forecastin

5.4.6 Support vector machine

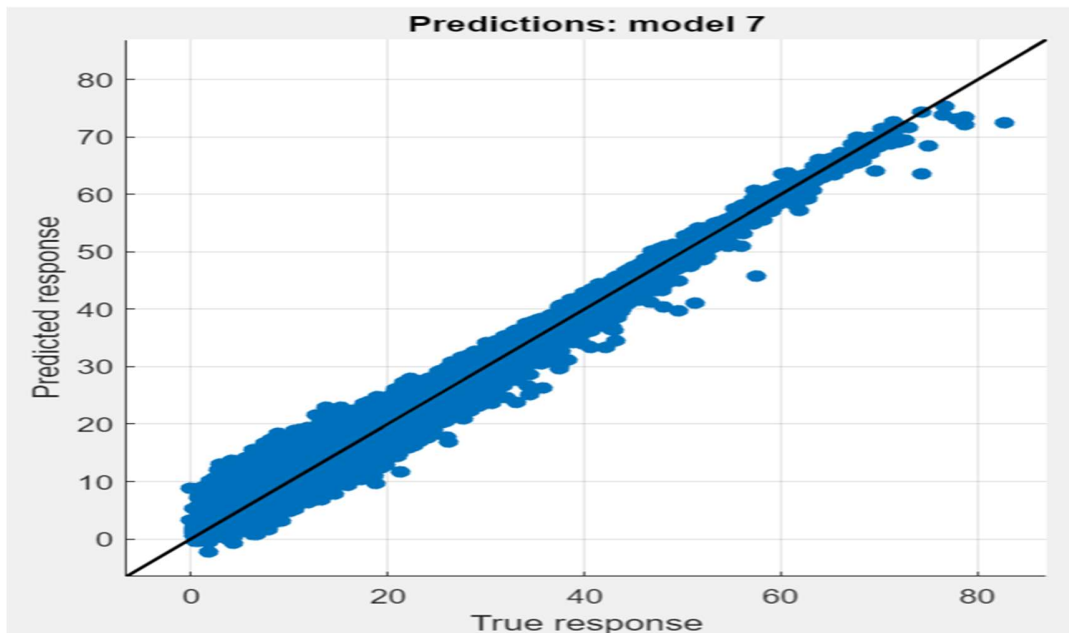


Figure 5.4.6.1 : SVM True vs predicted responses

The scatter plot comparing the true and predicted responses for the Support Vector Machine (SVM) model provides a strong visual confirmation of the model's accuracy. Most data points lie closely along the 45-degree diagonal line, indicating that the predicted wind speeds align well with the actual observed values. This close clustering shows that the model has learned the underlying data patterns effectively.

Although minor deviations are visible, especially at the extreme ends of the prediction range, these are relatively limited and do not significantly impact the overall prediction quality. The model handles mid-range values particularly well, where most of the data is concentrated.

This graphical analysis is consistent with the model's favorable error metrics, confirming that the SVM model is a reliable and consistent predictor for wind speed estimation. It maintains a solid balance between precision and generalization.

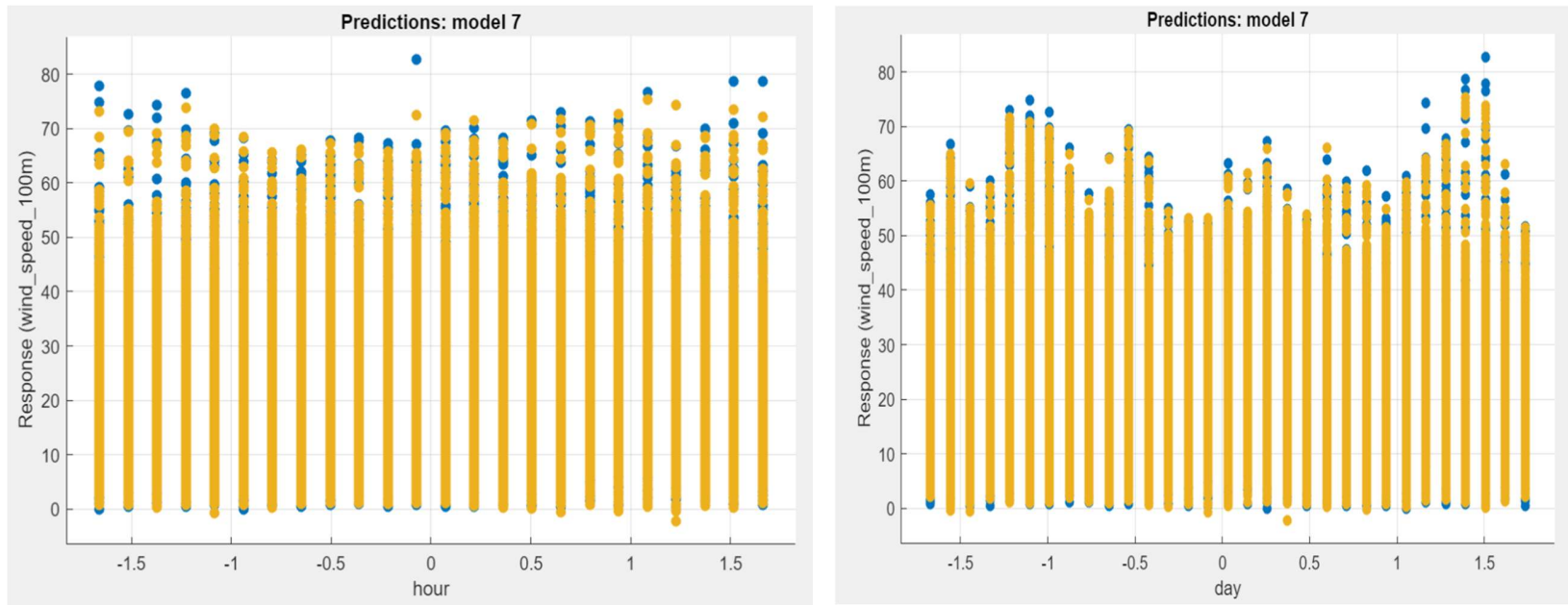


Figure 5.4.6.2 : Wind speed 100 m Response vs wind speed hourly and daily

The hourly and daily prediction plots provide a clear visual representation of the Support Vector Machine (SVM) model's performance over time. In both plots, the actual wind speeds (blue) and the predicted values (orange) show a strong level of alignment, particularly within the most frequent wind speed ranges. In the hourly plot, the SVM model tracks the short-term variations in wind speed with reasonable consistency, though minor gaps between predicted and true values appear at certain hours. In the **daily plot**, the model continues to follow broader trends effectively, maintaining a close approximation of the observed fluctuations across different days. These patterns align with the model's error results, which reflect solid predictive accuracy and low deviation from true values. The performance remains stable across both hourly and daily scales, indicating the model's capability to generalize well over different time intervals.

5.5 Models Results

This section presents a comparative evaluation of six different regression models used for predicting wind speed. The models were assessed using three standard performance metrics: Root Mean Squared Error (RMSE), R-squared (R^2), and Mean Absolute Error (MAE). These metrics provide insight into each model's predictive accuracy, error rate, and ability to explain variance in the observed data. As shown in the table, advanced models such as the Optimizable Tree and Ensemble Methods outperformed simpler linear models, indicating that complex, nonlinear patterns in the data are better captured by more sophisticated algorithms. This analysis helps identify the most suitable model for reliable and accurate wind speed forecasting.

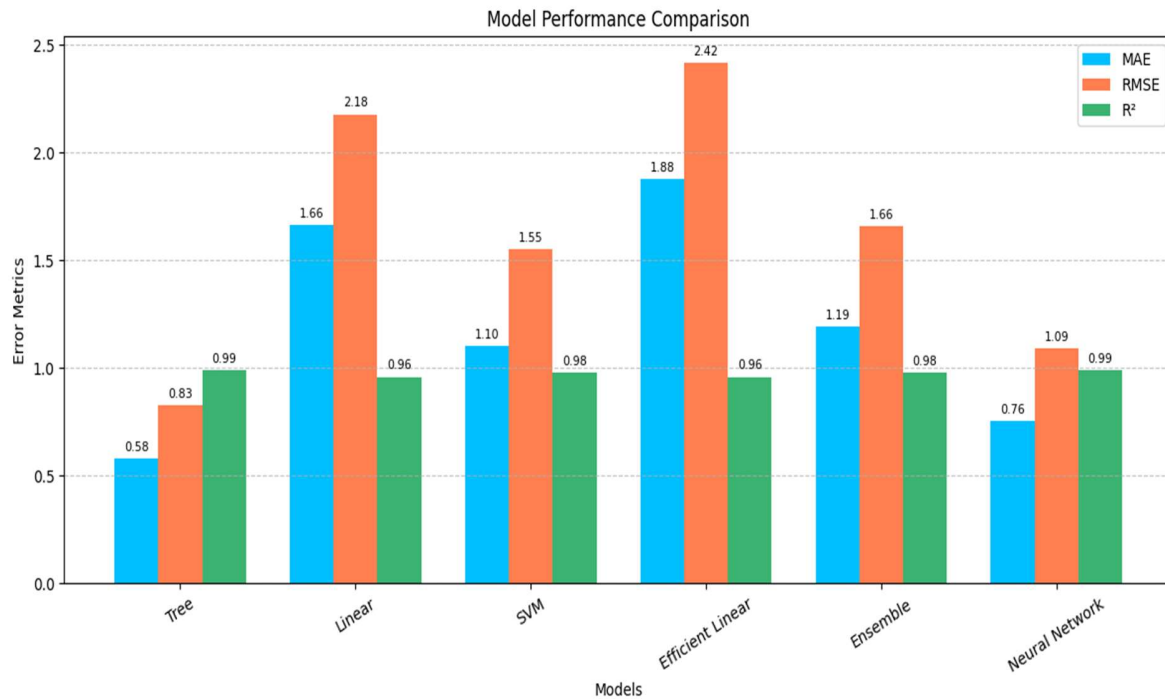


Figure 5.5 : Error results

This figure presents a graphical representation of the error metric and predictive accuracy values for the training of six models. Each model is depicted with its error metrics, including Root Mean Squared Error (RMSE), R-squared (R^2), and Mean Absolute Error (MAE).

5.5.1. Model Evaluation and Comparison

In this study, several machine learning models were trained and evaluated to predict wind speed using meteorological features. The models were assessed based on three key performance metrics: Root Mean Square Error (RMSE), Mean Absolute Error (MAE), and Coefficient of Determination (R^2). The results are summarized and discussed below.

1. Optimizable Tree Model

The optimizable tree model is a type of decision tree algorithm that benefits from automated hyperparameter tuning. Using optimization techniques such as Bayesian optimization, the model iteratively adjusts parameters including tree depth, split criteria, and minimum leaf size to improve prediction accuracy. This model is known for its interpretability and flexibility in handling both linear and non-linear data distributions. In this study, it achieved the best performance across all evaluated metrics, with an RMSE of **0.8276**, an MAE of **0.5826**, and an R^2 of **0.99**. These results indicate a high degree of accuracy and reliability, making the optimizable tree model the most effective approach for the given wind speed prediction task.

2. Linear Regression Model

Linear regression is a classical statistical technique used to model the linear relationship between dependent and independent variables. While it offers simplicity and interpretability, it assumes a linear relationship that may not hold in real-world datasets involving environmental variables. In this application, the linear regression model demonstrated relatively poor performance, with an RMSE of **2.1792** and an MAE of **1.6636**, despite achieving an R^2 of **0.96**. These results suggest that the model failed to adequately capture the underlying complexity and variability of the data.

3. Efficient Linear Model

The efficient linear model typically refers to linear regression variants optimized for speed and scalability, such as stochastic gradient descent regression or regularized regressors (e.g., Ridge or Lasso). These models are computationally efficient and suitable for high-dimensional datasets. However, similar to the standard linear regression model, they are limited in capturing non-linear patterns. The model yielded the **highest error values** among all tested algorithms, with an RMSE of **2.4193** and an MAE of **1.8765**, making it unsuitable for the present application.

4. Neural Network Model

Artificial neural networks (ANNs) are inspired by the human brain and consist of interconnected nodes (neurons) capable of learning complex, non-linear relationships through backpropagation. Although they require more computational resources and careful tuning, neural networks can generalize well in scenarios with complex data distributions. In this study, the neural network model produced an RMSE of **1.0932**, an MAE of **0.7561**, and an R^2 of **0.99**, ranking it as one of the top-performing models. Its ability to approximate intricate patterns makes it particularly effective for wind speed forecasting.

5. Ensemble Methods Model

Ensemble learning methods, including techniques such as bagging (e.g., Random Forest) and boosting (e.g., Gradient Boosting Machines), combine the predictions of multiple base learners to reduce variance and improve accuracy. These models are known for their robustness and generalization ability. The ensemble methods model in this study achieved an RMSE of **1.6593**, an MAE of **1.1942**, and an R^2 of **0.98**. While effective, it was slightly outperformed by the neural network and optimizable tree models.

6. Support Vector Machine (SVM) Model

Support Vector Machines for regression (SVR) are capable of modeling linear and non-linear relationships by using kernel functions to transform input data into higher-dimensional spaces. They are particularly effective when the number of features is large relative to the number of observations. In this work, the SVM model achieved an RMSE of **1.5514**, an MAE of **1.1031**, and an R^2 of **0.98**, indicating satisfactory performance with a good balance between complexity and accuracy.

6. Conclusion

This study presented a comprehensive approach to wind speed prediction using artificial intelligence (AI) techniques. The process began with essential data preparation steps, including the careful selection, cleaning, and normalization of meteorological data. These steps ensured that the dataset was accurate, consistent, and well-suited for machine learning analysis.

Following preprocessing, multiple machine learning models were developed and evaluated to identify the most accurate predictive method. The models included Decision Tree, Linear Regression, Efficient Linear Model, Neural Network, Ensemble Methods, and Support Vector Machine (SVM). Each model was trained on a carefully structured dataset and assessed using key performance metrics: Root Mean Squared Error (RMSE), Mean Absolute Error (MAE), and the R-squared (R^2) coefficient.

The results revealed varying levels of predictive performance across the models. The Optimizable Tree Model achieved the highest R^2 value of 0.99 and the lowest RMSE and MAE values, indicating excellent predictive capability. The Neural Network and Ensemble Methods models also showed strong performance, offering a good balance between accuracy and stability. Meanwhile, the SVM model demonstrated robust results, with an R^2 of 0.98 and relatively low error values, confirming its effectiveness for regression-based wind speed prediction.

Overall, this research highlights the importance of structured preprocessing and thorough model evaluation in AI-driven forecasting tasks. It emphasizes that model performance is highly dependent on the nature of the data and the algorithm's ability to generalize patterns. While several models performed well, the Decision Tree and SVM models stood out as top contenders, proving their suitability for future deployment in wind energy management systems. Further work will involve validating these models in real-world environments and integrating them into operational platforms to support smart grid planning and renewable energy forecasting.

7. Recommendations

To ensure the success and practical implementation of the AI models developed in this study for wind speed prediction, the following recommendations are proposed. These steps aim to maximize the project's impact on renewable energy forecasting, enhance the performance of microgrid systems, and support the development of reliable tools for wind energy management and sustainable energy planning.

1. Prioritize advanced model validation and tuning

- Conduct comprehensive cross-validation and hyperparameter tuning to further improve the Support Vector Machine (SVM) and Decision Tree models, which demonstrated the highest prediction accuracy.
- Evaluate alternative AI models or hybrid approaches to ensure robust performance across varying datasets and conditions.

2. Develop API integration for real-time applications

- Build a well-documented and secure API that integrates the chosen predictive model into microgrid or energy management platforms, allowing real-time wind speed forecasting.
- Ensure that the API supports scalability and can be adapted to various renewable energy systems.

3. Design an interactive monitoring dashboard

- Create a user-friendly dashboard for operators and decision-makers, enabling real-time visualization of wind speed forecasts and system performance metrics.
- Include trend analysis and alerts to support proactive grid management.

4. Collaborate with industry stakeholders

- Partner with energy companies or grid operators to implement pilot projects that test the AI models under real-world conditions.
- Use operational feedback to refine the models and enhance their reliability in practical scenarios.

5. Establish continuous model evaluation and updating mechanisms

- Monitor prediction accuracy regularly to detect any performance drift or degradation.
- Update models with fresh meteorological data to maintain adaptability to evolving environmental conditions.

6. Ensure transparency and promote further research

- Maintain detailed documentation of all processes, including data sources, feature selection, preprocessing steps, and model performance evaluation.
- Share findings with academic and industry communities to encourage collaboration, improvement, and the exploration of more innovative AI solutions in renewable energy forecasting.

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IEEE Code of Ethics



We, the members of the IEEE, in recognition of the importance of our technologies in affecting the quality of life throughout the world, and in accepting a personal obligation to our profession, its members and the communities we serve, do hereby commit ourselves to the highest ethical and professional conduct and agree:

1. to accept responsibility in making engineering decisions consistent with the safety, health and welfare of the public, and to disclose promptly factors that might endanger the public or the environment;
2. to avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
3. to be honest and realistic in stating claims or estimates based on available data;
4. to reject bribery in all its forms;
5. to improve the understanding of technology, its appropriate application, and potential consequences;
6. to maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
7. to seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, and to credit properly the contributions of others;
8. to treat fairly all persons regardless of such factors as race, religion, gender, disability, age, or national origin;
9. to avoid injuring others, their property, reputation, or employment by false or malicious action;
10. to assist colleagues and co-workers in their professional development and to support them in following this code of ethics.

Approved by the IEEE Board of Directors
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